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Certainty amid uncertainty: Relationship between macroeconomic uncertainty and individual expectations☆,☆☆

Giulia Piccillo [a,b], Poramapa Poonpakdee [c,*]

[a] Maastricht University, Netherlands
[b] CESifo, Munich, Germany
[c] Bank of Thailand, Thailand

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ABSTRACT

We study how individual income expectations adjust during periods of high macroeconomic uncertainty, using survey data from professional forecasters and households. We find that mean expectations decrease disproportionally with rising macroeconomic uncertainty, as predicted by most economics and finance models. However, the relationship between macroeconomic uncertainty and individual subjective uncertainty is more nuanced. While there is a positive correlation among professional forecasters, it is not the case among households. In fact, for households, the correlation is often negative, contrary to what is typically predicted. In addition, subjective uncertainty shows significant persistence, even after controlling for macroeconomic and individual factors. We demonstrate that our empirical results can be reconciled with Max-Min expected utility and smooth ambiguity preferences. This has implications for economic and financial research seeking to study periods of high macroeconomic uncertainty.

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1. Introduction

A growing body of literature examines the effects of macroeconomic uncertainty on the economy. Agents in these models perceive uncertainty either as risk, modeled as an increase in the standard deviation of time-varying shocks perceived by the agents (Bloom, 2009; Born et al., 2018; Born & Pfeifer, 2021; Fernández-Villaverde & Guerrón-Quintana, 2020), or as ambiguity, modeled as a wider range of beliefs (Altug et al., 2020; Ilut & Schneider, 2014, 2022; Marinacci, 2015; Ozoguz, 2009; Piccillo & Poonpakdee, 2023). While these two concepts carry different implications, both lead to a broader range of expected outcomes and to more pessimistic expectations in equilibrium. In other words, during periods of high macroeconomic uncertainty, people are assumed to be more uncertain about their expectations and to anticipate pessimistic outcomes, which in turn hinders economic growth. However, there is scarce evidence on how individual expectations of ordinary economic agents, such as households, actually adjust to macroeconomic uncertainty and on how these relationships align with decision-making theories. These are essential questions to answer if we are to understand the impact of macroeconomic uncertainty on the economy.

This paper examines the relationship between macroeconomic uncertainty and individual expectations using individual survey data from professional forecasters (the

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* Correspondence to: Bank of Thailand, 273 Samsen Road, Watsamphraya, Phra Nakhon District, Bangkok, 10200, Thailand.

E-mail addresses: g.piccillo@maastrichtuniversity.nl (G. Piccillo), poramapp@bot.or.th (P. Poonpakdee).

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Survey of Professional Forecasters, or SPF) and households (the US Survey of Consumer Expectations, or SCE). Specifically, we focus on GDP growth expectations from the SPF and income growth expectations from the SCE. We use income expectations rather than inflation because the straightforward interpretation of optimism and pessimism associated with expectations is a primary source of business cycle fluctuations (Milani, 2011). We quantify expectations by the first and second moments of the subjective forecast distributions provided by survey respondents. Throughout the paper, we refer to the first and second moments of individual subjective forecast distributions, respectively, as mean expectations and subjective uncertainty.[1] In our analysis, we study these individual variables in relation to three indices of macroeconomic uncertainty: the Economic Policy Uncertainty index (Baker et al., 2016), the macroeconomic uncertainty index by Jurado et al. (2015), and professional forecasters' disagreement.

We find a negative correlation between mean expectations and macroeconomic uncertainty among both professional forecasters and households, even after controlling for economic conditions and time-fixed effects. This result remains robust across different measures of mean expectations and macroeconomic uncertainty. It suggests that people become more pessimistic when macroeconomic uncertainty is higher. This finding is consistent with macroeconomic models that predict pessimistic expectations in times of uncertainty (Altug et al., 2020; Bhandari et al., 2025; Bloom, 2014; Fernández-Villaverde & Guerrón-Quintana, 2020; Ilut & Schneider, 2014, 2022; Ozoguz, 2009; Pellegrino et al., 2023; Piccillo & Poonpakdee, 2023).

In contrast, the relationship between subjective uncertainty and macroeconomic uncertainty differs sharply between the two groups. For professional forecasters, subjective uncertainty rises with macroeconomic uncertainty, consistent with theoretical assumptions that uncertainty widens the range of expected outcomes. For households, however, this significant positive correlation does not hold and is often negative. The household results remain robust across 52 empirical specifications using several measures of subjective and macroeconomic uncertainty, as illustrated in Fig. 1. Moreover, we find that subjective uncertainty exhibits significant persistence over time, suggesting that the distribution of beliefs adjusts gradually and may not respond immediately to changes in macroeconomic conditions or policy communication.

The divergence in subjective uncertainty between the two groups stems from how they adjust their forecast distributions in response to higher macroeconomic uncertainty. Professional forecasters tend to decrease the pessimistic tail of their forecast histogram, exhibiting more pessimistic expectations and greater subjective uncertainty. In contrast, households truncate the optimistic tail of their distribution while leaving the level of their lower bound largely unchanged. This effectively compresses their forecast histogram toward pessimistic outcomes, yielding smaller subjective uncertainty. Thus, although both groups become more pessimistic under high macroeconomic uncertainty, the underlying mechanisms differ fundamentally. This questions the empirical validity of the simple assumption that macroeconomic uncertainty is associated with an increased range of potential (perceived) outcomes at the individual level.

Finally, we reconcile these empirical findings with decision-making theories under uncertainty. We conceptualize macroeconomic uncertainty as a combination of risk and ambiguity, and show that our results align with Max-Min expected utility with multiple priors (Gilboa & Schmeidler, 1989) and smooth ambiguity preferences (Klibanoff et al., 2005). In particular, if ambiguity increases and people facing different scenarios of the world move towards Max-Min expected utility (or if they hold smooth ambiguity preferences), they will have stronger beliefs toward the pessimistic scenarios, reducing the perceived probability of the more optimistic ones. This mechanism aligns with the observed decrease in households' individual subjective uncertainty during periods of rising uncertainty. We also discuss alternative mechanisms, such as rational inattention, that can generate a non-monotonic correlation between macroeconomic uncertainty and individual subjective uncertainty.

This study offers three contributions. First, we add evidence on survey-based measures of subjective uncertainty, building upon existing research on professional forecasters (Clements, 2014; Giordani & Söderlind, 2003; Glas & Hartmann, 2016; Manzan, 2021) and firms (Altig et al., 2020; Bachmann et al.; Easaw & Grimme, 2024; Kumar et al., 2023). Our empirical approach to measuring households' subjective uncertainty is similar to that of Glas (2020), and we further adopt the alternative measurements in Bachmann et al., Altig et al. (2020), and Krüger and Pavlova (2024) in our robustness checks. Second, our results provide direct evidence on the relationship between macroeconomic uncertainty and individual expectations, in a context that is comparable to the assumptions of standard macroeconomic and financial models. Finally, we provide an analytical framework that bridges the gap between decision-making theories under uncertainty and survey-based measures of beliefs.

This paper is structured as follows. Sections 2 and 3 describe the survey data used and outline our empirical approach. Section 4 presents the main results and robustness tests. Section 5 discusses the findings in light of decision-making theories under uncertainty. Section 6 provides a brief conclusion.

2. Data

We use expectation data from two survey datasets: the Survey of Consumer Expectations (SCE) of US households and the US Survey of Professional Forecasters (SPF). This section describes these datasets.

Households - The Survey of Consumer Expectations (SCE) provides US household expectation data and is administered by the Federal Reserve Bank of New York. The

[1] This does not imply that macroeconomic uncertainty is objective uncertainty. Subjective uncertainty is a common term referring to the second moment of the forecast distribution, as in Altig et al. (2020), Enke and Graeber (2023), Fermand et al. (2024), Giordani and Söderlind (2003), Glas and Hartmann (2016). We follow the same practice.

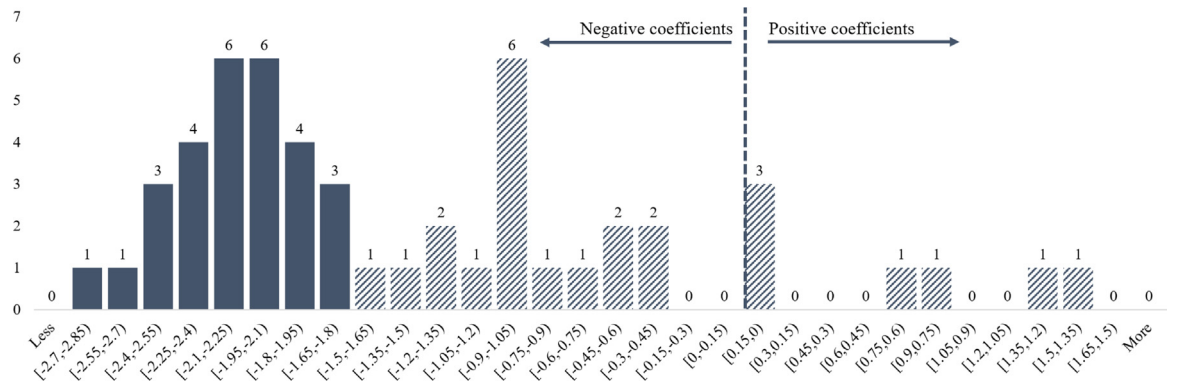


Fig. 1. T-statistics of the coefficients linking macroeconomic uncertainty and household subjective uncertainty.
Note: This histogram presents an overview of the t-statistics for the coefficients between macroeconomic uncertainty and households' subjective uncertainty. The data are based on the 52 distinct empirical specifications. A solid bar indicates that the coefficient is significant at the 10% level, while a lightly shaded bar indicates that the coefficient is not significant.

monthly sample spans from 2013M6 to 2024M7. The survey provides a point forecast for the next 12 months' gross personal income growth, so the variable has a 12-month fixed-horizon forecast. The income growth point forecast is used as the mean expectations.

Moreover, the SCE asks for the probability that next-12-month income growth falls into 12 interval bins, ranging from more than 12% to less than -12%. The related questionnaire is "Suppose again that, 12 months from now, you are working in the same main job...Your earnings on this job, before taxes and deductions, will have increased by 0% to 2% (bin 6) _____ percent chance". These probabilistic bins form individual subjective forecast distributions to calculate subjective uncertainty. The number of observations is around 96,000, with a cross-section of 15,000 respondents, and the average time series per individual is 6 months. See the details of questionnaires and summary statistics in Appendix A.1.

Professional forecasters - We use the Survey of US Professional Forecasters collected by the Federal Reserve Bank of Philadelphia. Our sample is quarterly, ranging from 1992Q1 to 2025Q1. For SPF, the mean expectations are the forecasts of the next-calendar-year real GDP *growth*. Originally, the survey provides the point forecast of the next calendar year real GDP *level*, which is not comparable to the income growth forecasts in the households data. Therefore, we calculate the growth forecast by dividing next year's GDP by this year's forecast GDP. For example, if the survey was conducted in 2010, the forecast is based on the 2011 annual growth rate. This means the variable has a changing horizon that becomes shorter as we move toward the end of the year.

We use real GDP growth expectations and fixed-event forecasts—rather than fixed-horizon forecasts as in the household survey—because the SPF provides subjective probability distributions for this variable only. To form the subjective probability distribution, we use the questionnaires to estimate the probability that the next calendar year's real GDP growth falls into the interval bin from less than −2% to more than 6%. The interval bins were expanded in 2009Q2 and 2020Q2, and these technical changes affect the level of subjective uncertainty we need to control for in our regressions. The number of observations is around 4900, with a cross-section of 187 respondents, and each individual has an average time series of around 27 quarters. The details of the questionnaires and summary statistics are in Appendix A.2.

3. Empirical methods

The main question this paper addresses is how individual expectations respond to macroeconomic uncertainty. The standard approach is to associate a higher degree of uncertainty with a wider range of potential outcomes perceived by agents, leading to pessimistic expectations. Our dependent variables are mean expectations of income or GDP, acting as a proxy for pessimistic/optimistic expectations, and subjective uncertainty, serving as a proxy for the range of potential outcomes. The following sections discuss the dependent and independent variables and the identification strategy.

3.1. Dependent variables

Our study uses a variety of measurement methods to assess the robustness of the results. In the main analysis, we use the first moment as the measure of mean expectations and the second moment as the measure of subjective uncertainty.

Mean expectations – We obtain the mean expectation of income or GDP growth using three approaches. First, it is the point forecast directly provided by the survey respondents. Second, we apply the mass-at-midpoint (MAM) method, which assumes that the probability mass is located at the midpoint of each forecast bin. The MAM mean expectation is computed as:

$$\text{Mean expectations (MAM)}_{i,t} = \sum_{b} \left(\text{Probability}_{b,i,t} \times \text{Midpoint}_{b,i,t}\right)$$

where b is the bin identifier, i is the cross-sectional identifier, and t is time.

Third, we fit a generalized beta distribution (GBD) to the subjective histogram provided by the respondents and obtain the first moment of the fitted GBD as the mean expectations. The SCE reports the GBD mean expectations following Armantier et al. (2017). For the SPF, we adopt programming code from Tao (2021) to measure the GBD mean expectations.[2] The correlations between these three measurements are significantly positive, ranging from 0.84 to 0.99 for SPF and from 0.64 to 0.96 for SCE.

Subjective uncertainty – We compute subjective uncertainty, which is the perceived uncertainty of an individual about their economic forecast. This is usually measured as the second moment of a subjective forecast distribution (Altig et al., 2020; Coibion et al., 2024; Enke & Graeber, 2023; Fermand et al., 2024; Giordani & Söderlind, 2003; Glas & Hartmann, 2016). The two main ways of assessing the second moment are MAM and GBD.

First, the MAM measure of subjective uncertainty is defined as:

$$\text{Subj. uncertainty (MAM)}_{i,t} = \sqrt{\sum_{b} \left[\text{Probability}_{b,i,t} \times \left(\text{Midpoint}_{b,i,t} - \text{Mean expectations}_{i,t} \right)^2 \right]}$$

where b is the bin identifier, i is the cross-sectional identifier, and t is time. The mean expectation used here is calculated using the mass-at-midpoint method.

Second, the GBD subjective uncertainty is the second moment of generalized beta distributions fitted to the respondents' subjective forecast distributions. This method was proposed by Engelberg et al. (2009) and has been adopted, for instance, by Clements (2014) and Glas and Hartmann (2016). The SCE provides the fitted subjective uncertainty following Armantier et al. (2017). For the SPF, we use the programming code from Tao (2021), as for the GBD mean expectations, to measure the GBD subjective uncertainty. The correlations between the two measures are high, at 0.92 for the SPF and 0.86 for the SCE.

The subjective uncertainty measures of MAM and GBD are characterized as second moments in a distribution and, as such, essentially require a histogram with a standardized shape, including a unimodal form and no gaps between bins. However, these properties may not always be guaranteed in the histograms provided by respondents, particularly in household surveys. To address this limitation, we incorporate two additional measures that are less sensitive to assumptions about histogram shape. First, we compute the Expected Ranked Probability Score (ERPS) proposed by Krüger and Pavlova (2024), which is based on the concept of entropy.[3] This method requires consistent bin definitions across time. Since the SPF bin format changed twice over our sample period, we exclude ERPS from the SPF regressions. Second, we use the span—the difference between the upper and lower bound forecasts—as a proxy for subjective uncertainty, following Bachmann et al.. This measure captures the width of the subjective histogram and reflects the range of possible outcomes respondents consider.

For clarity, the main text reports results using respondents' mean expectations and subjective uncertainty from the MAM method. Results using the alternative measures are presented in Appendix B as robustness checks.

3.2. Independent variables

We use three macroeconomic uncertainty indices as our independent variables to validate the reliability of our findings. The first two indices are the Economic Policy Uncertainty (EPU) index by Baker et al. (2016) and the 1-month-ahead Macroeconomic Uncertainty (JU) index by Jurado et al. (2015). The EPU index is the most commonly used measure in the empirical literature on macroeconomic uncertainty. It is constructed from the frequency of terms related to economic and policy uncertainty appearing in major newspapers, is available for a wide range of countries, and is published daily. The second index, the 1-month-ahead Macroeconomic Uncertainty (JU) index, is based on the unpredictable components of a large set of macroeconomic time series. It is forward-looking and grounded in a strong econometric framework, making it a benchmark in recent literature on macroeconomic uncertainty.

Lastly, we also consider the forecast disagreement for next year's GDP growth, calculated as the cross-sectional standard deviation across the SPF forecasts. The use of forecast disagreement as a proxy for macroeconomic uncertainty remains debated. Some studies employ it directly (Coibion et al., 2024; Dovern et al., 2012), while others caution against its use (Born et al., 2023; Glas, 2020; Lahiri & Sheng, 2010). For example, Born et al. (2023) show that agents respond differently to forecast disagreement and to the Jurado et al. (2015) uncertainty index. While greater uncertainty raises the value of new information and amplifies agents' reactions to news, higher disagreement lowers the perceived reliability of information and dampens their responsiveness.

Overall, the three indices are positively correlated: EPU–JU (0.51), JU–disagreement (0.55), and EPU–disagreement (0.39) over the period 1992Q1–2025Q1, suggesting that they generally capture similar uncertainty shocks.

3.3. Regression setup

We aim to find empirical evidence on whether the survey respondents are more pessimistic and more uncertain during periods of high macroeconomic uncertainty. Specifically, we investigate the correlation between individual expectations and macroeconomic uncertainty. To do this, we conduct pooled ordinary least squares (OLS) regressions of individual expectations on macroeconomic

[2] We assume that the subjective histogram is unimodal, i.e., the two parameters of the generalized beta distributions (a and b) are greater than one, as done in Armantier et al. (2017) and Engelberg et al. (2009).

[3] Higher entropy in the forecast distribution implies greater likelihood of forecast errors, and thus higher subjective uncertainty. The ERPS subjective uncertainty is calculated as $\sum_{bin}(1 - P_{bin})$ where P_{bin} is the probability assigned to a given bin.

uncertainty indices, accounting for clustered standard errors at both the individual and time levels.[4] Our core models take the following forms.

Professional forecasters:

$$\begin{aligned} Y_{i,t} &= \alpha_0 + \alpha_1 Y_{i,t-1} + \alpha_2 \text{Macro uncertainty}_t \\ &\quad + \alpha_3 \text{Real GDP growth}_{t-1} + \sum_{j=2}^{3} \alpha_{4,j} \text{Survey format dummy}_{j,t} \\ &\quad + \sum_{j=2}^{4} \alpha_{5,j} \text{Quarter dummy}_{j,t} + \sum_{j=1993}^{2025} \alpha_{6,j} \text{Year dummy}_{j,t} \\ &\quad + \sum_{i=2}^{N} \alpha_{7,i} \text{Individual dummy}_i + \epsilon^{\alpha}_{i,t} \end{aligned}$$

Households:

$$\begin{aligned} Y_{i,t} &= \beta_0 + \beta_1 Y_{i,t-1} + \beta_2 \text{Macro uncertainty}_t \\ &\quad + \beta_3 \text{Real GDP growth}_{t-1} + \beta_4 \text{Employment rate}_{t-1} \\ &\quad + \sum_{j=2}^{12} \beta_{5,j} \text{Month dummy}_{j,t} + \sum_{j=2014}^{2024} \beta_{6,j} \text{Year dummy}_{j,t} \\ &\quad + \sum_{i=2}^{N} \beta_{7,i} \text{Individual dummy}_i + \epsilon^{\beta}_{i,t} \end{aligned}$$

where $Y_{i,t}$ is mean expectations or subjective uncertainty. $Y_{i,t-1}$ is the lagged dependent variable, which controls for past momentum and prior beliefs. We perform unit-root tests with one lag for all dependent variables, and all tests reject the null hypothesis of a unit root. Macro uncertainty is the macroeconomic uncertainty index on a log scale, which is the variable of interest. We account for business cycles by incorporating real-time GDP growth from the preceding quarter. Since the household data are monthly, we assume the GDP growth is the same for all three months in the current quarter. Additionally, in the household data regression analysis, real-time monthly employment rates are included as a control for monthly business cycles. Finally, there are time and individual fixed effects. We attempted to control for COVID-periods by introducing 2020–2022 time dummies. However, these are collinear with the year fixed effects, thus the COVID dummies are omitted.

Controlling for the quarterly fixed effects in SPF is essential because its forecasting variable, the next-calendar-year real GDP growth, has a shorter horizon when moving to the next quarter. This increases forecasters' confidence (Clements, 2014; Giordani & Söderlind, 2003; Manzan, 2021) and reduces the cost of information, thereby improving forecast performance (Carroll, 2003; Lamla & Lein, 2014). Moreover, we control for the changes in the survey format related to the subjective forecasting distributions by including time dummies for 2009Q2–2020Q1 and 2020Q2–2023Q3.

[4] We use two-way clustered standard errors (by individual and time) to account for potential correlation in the error terms both within individuals over time and across individuals at a given point in time. This is appropriate given the panel structure and the presence of time-series controls. In specifications where the number of clusters in one dimension (particularly time) is insufficient to ensure reliable inference, we instead report standard errors clustered at the individual level only.

Our analysis is designed to capture correlation rather than to identify causal effects. Establishing causality would require exogenous variation in macroeconomic uncertainty or an alternative identification strategy, which is beyond the scope of this paper. Nevertheless, in Section 4.3, we examine the temporal ordering of shocks by using daily Economic Policy Uncertainty data to ensure that movements in uncertainty precede respondents' survey completion dates.

In the subjective-uncertainty regressions, we include both lagged subjective uncertainty and professional forecasters' disagreement. Following Lahiri and Sheng (2010), these variables can represent components of aggregate forecast uncertainty when measured within the same group of forecasters.[5] Including both may therefore introduce collinearity, as they capture similar information, particularly in the professional forecaster sample. For households, however, professional forecasters' disagreement serves as an external indicator of macroeconomic uncertainty rather than a component of households' aggregate forecast uncertainty. The Lahiri–Sheng decomposition does not apply in this case, so we retain the SPF disagreement in the subjective-uncertainty regressions for completeness.

4. Results

In this section, we present our main results. First, we assess whether people are more pessimistic in times of high macroeconomic uncertainty by examining mean expectations. Second, we investigate whether people are more uncertain in periods of high macroeconomic uncertainty by examining subjective uncertainty. Finally, to deepen our understanding of these two findings, we examine the temporal linkage between macroeconomic uncertainty and individual expectations and analyze changes in the shape of the forecast histogram.

4.1. Mean expectations

This section explores the relationship between macroeconomic uncertainty and the mean expectations of GDP and income growth for professional forecasters and households. Table 1 presents the baseline regression results, where Columns (1)–(3) report estimates for professional forecasters and Columns (4)–(6) for households. In all models, the dependent variable is the mean expectation directly reported by survey respondents—next-calendar-year GDP growth for professional forecasters and next-12-month income growth for households.

Across both groups, we find that higher macroeconomic uncertainty is significantly associated with lower mean expectations. For instance, a 1% increase in the Economic Policy Uncertainty index is linked to a 0.817% reduction in annual GDP growth expectations among professional forecasters—roughly one-third of a typical yearly contraction—and a 0.0982% reduction in income growth

[5] That is, both disagreement and subjective uncertainty are derived from the same group of forecasters.

Table 1
Mean expectations of professional forecasters and households.

	Professional forecasters			Households		
	(1)	(2)	(3)	(4)	(5)	(6)
Log Economic Policy Uncertainty	−0.817***			−0.0982**		
	(0.159)			(0.0378)		
Log Jurado et al. index		−2.132***			−0.549**	
		(0.805)			(0.222)	
Log Forecasters disagreement			−0.463**			−0.176***
			(0.214)			(0.0566)
Lagged real-time GDP growth	0.00917**	0.00556	0.00912	0.00300**	0.00144	0.00118
	(0.00442)	(0.00491)	(0.00601)	(0.00118)	(0.00123)	(0.00123)
Lagged real-time employment rate				0.00195**	0.00159*	0.00154*
				(0.000983)	(0.000889)	(0.000792)
Being unemployed				−0.407***	−0.398***	−0.391***
				(0.147)	(0.147)	(0.147)
Lagged dependent variable	0.466***	0.451***	0.461***	0.0638***	0.0636***	0.0636***
	(0.0444)	(0.0460)	(0.0442)	(0.00852)	(0.00853)	(0.00852)
Constant	5.517***	0.542	1.507***	5.178***	4.417***	4.486***
	(0.782)	(0.465)	(0.223)	(0.331)	(0.252)	(0.249)
Number of observations	4025	4025	4025	68 702	68 702	68 702
Number of individuals	150	150	150	10 037	10 037	10 037
Adjusted R-squared	0.63	0.63	0.62	0.60	0.60	0.60
Year-Quarter/Year-Month FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Survey format FE	Y	Y	Y			

Note: The standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals (professional forecasters and households). The dependent variable is the mean expectation directly provided by the respondents. Year-quarter fixed effects for professional forecasters are implemented using year and quarter dummies. For households, we use year and month dummies. Individual fixed effects are dummies for each individual. See Appendix B.1 Tables 9 and 10 for robustness checks.

expectations among households. These effects remain significant even after controlling for real-time macroeconomic conditions (GDP growth and employment rate), individual employment status, lagged expectations, and fixed effects for time, individuals, and survey format.

The setup of Table 1 accounts for individuals to form rationally negative forecasts during periods of economic contraction, as this effect is controlled for by lagged GDP growth. The fact that business cycle variables do not fully account for the correlation between macroeconomic uncertainty and mean expectations suggests that individuals tend to be overly pessimistic in times of heightened uncertainty, regardless of whether actual economic conditions justify such pessimism.

The negative relationship between macroeconomic uncertainty and mean expectations is robust across alternative measures and model specifications. Appendix B.1 and Tables 9 and 10 present an additional 24 sets of robustness checks that incorporate alternative measures of mean expectations. In total, we estimate 27 specifications for each group, with consistently negative coefficients on macroeconomic uncertainty, though not all are statistically significant. This robustness across 27 specifications reinforces that pessimism is a systematic response to macroeconomic uncertainty, not a result of model choice.

The estimated coefficients on the control variables are broadly consistent with theoretical expectations. Macroeconomic indicators, such as real-time GDP growth and employment rates, have a positive effect on mean expectations for both groups, suggesting that respondents form their forecasts in response to economic conditions. Among households, being unemployed significantly reduces income growth expectations by approximately 0.391% to 0.407% or about one-tenth of their average expected income growth of 3.43%. The lagged dependent variable is highly significant across all specifications, reflecting persistence in expectations over time. These findings suggest that both macroeconomic conditions and individual labor market status play important roles in shaping income expectations.

Including individual fixed effects raises the adjusted R-squared of the household regressions from about 43% to 60% (Table 10), but yields no improvement for professional forecasters (Table 9). This suggests that personal characteristics play a greater role in shaping household expectations. Das et al. (2020) show that socioeconomic status influences expectation formation, with lower-SES individuals typically more pessimistic—a pattern more relevant for households than for professional forecasters, whose forecasts concern aggregate output rather than personal income.

To summarize, we find that people become increasingly pessimistic during periods of high macroeconomic uncertainty, as evidenced by the negative relationship between macroeconomic uncertainty and mean expectations. This aligns with the assumptions of most macroeconomic business cycle models.

4.2. Subjective uncertainty

This section explores the relationship between individual subjective uncertainty and macroeconomic uncertainty. Table 2 presents the baseline regression re-

Table 2
Subjective uncertainty of professional forecasters and households.

	Professional forecasters			Households		
	(1)	(2)	(3)	(4)	(5)	(6)
Log Economic Policy Uncertainty	0.0936**			−0.0351**		
	(0.0379)			(0.0174)		
Log Jurado et al. index		0.327			−0.194**	
		(0.198)			(0.0878)	
Log forecasters' disagreement			0.132***			−0.0104
			(0.0453)			(0.0187)
Lagged real-time GDP growth	0.00213	0.00286	0.00274	0.00104*	0.000485	0.000995
	(0.00414)	(0.00391)	(0.00396)	(0.000606)	(0.000619)	(0.000685)
Lagged real-time employment rate				−0.000128	−0.000255	−0.0000373
				(0.000420)	(0.000380)	(0.000468)
Being unemployed				−0.0284	−0.0247	−0.0283
				(0.0710)	(0.0709)	(0.0709)
Lagged dependent variable	0.457***	0.460***	0.462***	0.0732***	0.0731***	0.0732***
	(0.0529)	(0.0514)	(0.0519)	(0.0107)	(0.0107)	(0.0107)
Constant	0.0774	0.695***	0.562***	3.814***	3.542***	3.625***
	(0.184)	(0.132)	(0.0669)	(0.149)	(0.120)	(0.117)
Number of observations	3761	3761	3761	64 350	64 350	64 350
Number of individuals	146	146	146	9336	9336	9336
Adjusted R-squared	0.75	0.75	0.76	0.74	0.74	0.74
Year-Quarter/Year-Month FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Survey format FE	Y	Y	Y			

Note: The standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals (professional forecasters and households). The dependent variable is the subjective uncertainty measured from the mass-at-midpoint approach. Year-quarter fixed effects for professional forecasters are implemented using year and quarter dummies. For households, we use year and month dummies. Individual fixed effects are dummies for each individual. See Appendix B.2 Tables 11 and 12 for robustness checks.

sults, with Columns (1)–(3) reporting estimates for professional forecasters and Columns (4)–(6) for households. In all models, the dependent variable is individual subjective uncertainty, constructed using the Mass-at-Midpoint (MAM) approach.

The results reveal a notable divergence in how professional forecasters and households respond to macroeconomic uncertainty. For professional forecasters, higher macroeconomic uncertainty is positively related to subjective uncertainty. For example, a 1% increase in the EPU index and forecast disagreement corresponds respectively to a 0.0936% and 0.132%[6] rise in subjective uncertainty. This positive relationship is consistent with Glas (2020) and Easaw and Grimme (2024), indicating that professional forecasters tend to express greater uncertainty about their projections during periods of heightened macroeconomic uncertainty.

In contrast, households respond in the opposite direction. Columns (4)–(6) show that greater macroeconomic uncertainty is significantly associated with lower subjective uncertainty. For instance, the coefficient on the Jurado et al. (2015) index is −0.194. This suggests that, unlike professional forecasters, households do not expand the range of their income growth forecasts when macroeconomic uncertainty is high—a puzzling result that warrants further investigation.

The divergent effects for professional forecasters and households are consistent with a broader set of specifications presented in Appendix B.2, Tables 11 and 12. Across 18 empirical specifications for professional forecasters, all coefficients on macroeconomic uncertainty are positive, with 9 of them statistically significant at the 10% level. For households, only 6 of 27 coefficients are positive, none are statistically significant, while the remaining 21 are negative, 9 of which are significant at conventional levels.

Unlike the regressions for mean expectations, the estimated coefficients on the control variables—such as real-time employment and unemployment status—are generally modest or statistically insignificant in the subjective uncertainty regressions. However, the lagged dependent variable remains highly significant across all specifications, reflecting persistence in subjective uncertainty over time. Subjective uncertainty among professional forecasters shows moderate persistence (a coefficient around 0.46), suggesting that forecasters gradually update their beliefs. By comparison, household uncertainty, which also displays significant persistence, has a lower coefficient (0.07), suggesting that they are more reactive to current events but short-lived. Such persistence has implications for how policymakers manage expectations and communicate under heightened macroeconomic uncertainty.

Being unemployed does not significantly impact households' subjective uncertainty about income growth. This result deserves further discussion. The share of unemployed respondents in our sample is very small (0.94%), since the income-expectation question refers specifically to income from *the same job* and therefore primarily targets employed individuals. Respondents classified as unemployed are those who report working and then not working between survey waves. These are normally temporary layoffs or individuals on leave—groups that

[6] As noted in Section 3, Lahiri and Sheng (2010) shows that both professional forecasters' disagreement and their subjective uncertainty may capture overlapping components of aggregate forecast uncertainty. This might bias the estimated coefficient on disagreement.

Table 3
Summary statistics of subjective uncertainty by personal instability.

Subjective uncertainty (MAM)	Mean	S.D.
Job tenure		
Work in main job $<$ 5 years	1.92	1.82
Work in main job $\geq$ 5 years	1.71	1.82
Education level		
No college degree	1.88	2.20
Have a college degree	1.79	1.60

typically expect to return to their previous positions. While these respondents report lower expected income, likely because they do not receive regular pay during this period, their subjective uncertainty about income remains largely unchanged, consistent with the view that they perceive their situation as temporary.

Including individual fixed effects substantially improves model fit. For households, the adjusted R-squared rises from 0.58 to 0.74 (Table 12); for professional forecasters, it increases from 0.70 to 0.75 (Table 11). This implies that individual characteristics contribute meaningfully to subjective uncertainty in both groups. This finding is in line with the results of Fermand et al. (2024) and Dovern (2024), who find that personal traits have an impact on households' subjective uncertainty regardless of the forecasting variables, and with Bachmann et al., who find that firms' characteristics are strongly associated with subjective uncertainty.

The contrasting results between households and professional forecasters are robust to two potential concerns: (i) the survey timeframe and (ii) households' numeracy levels. First, Table 13 shows that the positive and significant effect of macroeconomic uncertainty for professional forecasters persists even when restricting the sample to the same period as the household data (2013Q1–2025Q1). Second, as noted by Armantier et al. (2015), respondents with lower numeracy may provide expectations that are less consistent with economic theory. However, Table 14 shows that, regardless of numeracy level,[7] no significant positive effects are observed in the household models. In fact, some models still show negative and significant coefficients. These tests show that the different patterns observed for professional forecasters and households are robust to the extent to which this data can account for them.

Personal instability - The above findings suggest that households do not necessarily become more uncertain when faced with higher macroeconomic uncertainty and, in several cases, may actually become more certain. This raises the possibility that household responses are influenced by personal instability, which might shape their perceptions and responses to macroeconomic uncertainty.

To explore this possibility, we examine whether household characteristics linked to personal instability—such as shorter job tenure or lower educational attainment—are associated with higher subjective uncertainty. Table 3 shows that households with less than five years in their main job or without a college degree tend to report higher subjective uncertainty. This implies that households in unstable conditions are, on average, more uncertain about their income growth, which is consistent with the finding of Fermand et al. (2024).

Table 4 reports the results of the subjective uncertainty of households with different job tenures and levels of education, as depicted in Panels A and B, respectively. Specifically, Columns (1)–(3) show the results of households with less stable conditions, defined as those with a job tenure of less than 5 years or without a college degree. On the other hand, Columns (4)–(6) represent households in more stable conditions. Similar to the previous sections, results using alternative measures of subjective uncertainty are provided in Appendix B.2.

Across all specifications, none of the macroeconomic uncertainty coefficients are significantly positive. In Panel A, for both short- and long-tenure job groups, the coefficients for the EPU and Jurado et al. (2015) indices are negative, though the levels of statistical significance differ. For example, the Jurado et al. (2015) coefficient is -0.277 for households with job tenure of less than five years and -0.0923 for those with longer tenure. Panel B shows a similar pattern, with both indices again yielding negative coefficients. The disagreement measure has opposite signs across groups, but none are statistically significant. Despite these differences in job tenure and educational attainment, we find no robust evidence that personal instability meaningfully alters the relationship between household subjective uncertainty and macroeconomic uncertainty.

In summary, professional forecasters exhibit more dispersed forecast distributions during periods of high macroeconomic uncertainty, which is commonly modeled in economic and financial models as time-varying volatility or a wider range of prior beliefs. However, we find no evidence showing that households' subjective uncertainty increases with macroeconomic uncertainty. Most of the time, we find a negative relationship. This novel insight adds to the literature, indicating that the relationship between macroeconomic and individual subjective uncertainty may not be monotonic or simply positive.

4.3. Temporal link between macroeconomic uncertainty and individual expectations

At this point, readers may wonder why we have not explicitly addressed the issue of causality. While establishing a causal relationship is certainly desirable, doing so would require a more advanced identification strategy and likely a different dataset. For example, Coibion et al. (2024) use randomized information treatments in a

[7] The SCE assigns the level of numeracy based on the questions about probabilities and compound interest.

Table 4
Subjective uncertainty of households by job tenure and education.

Panel A: Job tenure	Main job < 5 years			Main job ≥ 5 years		
	(1)	(2)	(3)	(4)	(5)	(6)
Log Economic Policy Uncertainty	−0.0368			−0.0316		
	(0.0235)			(0.0247)		
Log Jurado et al. index		−0.277*			−0.0923	
		(0.140)			(0.105)	
Log forecasters' disagreement			0.00469			−0.0186
			(0.0294)			(0.0300)
Lagged real-time GDP growth	0.00202***	0.00119	0.00215**	−0.000200	−0.0000311	0.0000516
	(0.000871)	(0.000899)	(0.00101)	(0.000789)	(0.000811)	(0.000885)
Lagged real-time employment rate	−0.000507	−0.000736*	−0.000340	0.000525	0.000520	0.000562
	(0.000481)	(0.000430)	(0.000577)	(0.000510)	(0.000524)	(0.000548)
Being unemployed	−0.101	−0.0945	−0.101	−0.0145	−0.0135	−0.0144
	(0.115)	(0.114)	(0.115)	(0.0764)	(0.0766)	(0.0765)
Lagged dependent variable	0.0841***	0.0841***	0.0841***	0.0653***	0.0653***	0.0653***
	(0.0143)	(0.0143)	(0.0143)	(0.0150)	(0.0149)	(0.0150)
Constant	3.863***	3.541***	3.681***	3.625***	3.421***	3.446***
	(0.229)	(0.207)	(0.202)	(0.203)	(0.165)	(0.154)
Number of observations	27 958	27 958	27 958	34 028	34 028	34 028
Number of individuals	4059	4059	4059	4800	4800	4800
Adjusted R-squared	0.71	0.71	0.71	0.76	0.76	0.76
Year-Month FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Panel B: Education	Without college education			With college education		
	(1)	(2)	(3)	(4)	(5)	(6)
Log Economic Policy Uncertainty	−0.0863***			−0.00878		
	(0.0327)			(0.0199)		
Log Jurado et al. index		−0.175			−0.192**	
		(0.176)			(0.0927)	
Log forecasters' disagreement			−0.0510			0.00711
			(0.0444)			(0.0209)
Lagged real-time GDP growth	0.00430***	0.00392**	0.00389**	−0.000354	−0.000958*	−0.000254
	(0.00138)	(0.00152)	(0.00156)	(0.000522)	(0.000535)	(0.000601)
Lagged real-time employment rate	−0.000740	−0.000634	−0.000633	0.000128	−0.0000955	0.000193
	(0.000948)	(0.000975)	(0.00100)	(0.000345)	(0.000323)	(0.000387)
Being unemployed	−0.0649	−0.0618	−0.0616	−0.00108	0.00183	−0.00214
	(0.119)	(0.119)	(0.118)	(0.0820)	(0.0821)	(0.0817)
Lagged dependent variable	0.0724***	0.0727***	0.0727***	0.0713***	0.0712***	0.0713***
	(0.0154)	(0.0154)	(0.0154)	(0.0128)	(0.0127)	(0.0128)
Constant	4.678***	4.155***	4.190***	3.351***	3.215***	3.315***
	(0.260)	(0.222)	(0.210)	(0.180)	(0.146)	(0.146)
Number of observations	22 227	22 227	22 227	42 123	42 123	42 123
Number of individuals	3392	3392	3392	5944	5944	5944
Adjusted R-squared	0.76	0.76	0.76	0.72	0.72	0.72
Year-Month FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y

Note: The standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals (households). The dependent variable is the subjective uncertainty measured from the mass-at-midpoint approach. We use year and month dummies. Individual fixed effects are dummies for each individual. See Appendix B.2 Tables 16 and 17 for robustness checks.

European household survey to test the causal impact of macroeconomic uncertainty on expectations and spending behavior. Within our constraints, the most direct step we can take is to use the daily Economic Policy Uncertainty (EPU) index (Baker et al., 2016) to identify the level of macroeconomic uncertainty on the specific day when each respondent completed the survey. Because the EPU is based on newspaper coverage typically released in the morning, we can reasonably assume that the uncertainty signal precedes the respondent's survey submission. By matching each household's expectations to the daily EPU value on their response date, we establish a temporal ordering between macroeconomic uncertainty and individual expectations.

This approach helps mitigate concerns about reverse causality by ensuring the correct temporal ordering between macroeconomic uncertainty and survey responses. While it does not constitute a full causal identification strategy, it exploits quasi-exogenous timing variation in macroeconomic uncertainty. In particular, it supports the idea that daily fluctuations in macroeconomic uncertainty may influence how households form expectations. Unfortunately, this method cannot be extended to the SPF

data, as response dates are not recorded.[8] Despite this limitation, the household survey allows us to explore a more temporally precise link between macroeconomic uncertainty and individual expectations.

Table 5 presents the regression results using the daily EPU index, reinforcing the directional consistency of our main results. Columns (1)–(3) show that when the daily EPU is higher, the income growth expectations become lower, with coefficients ranging from −0.039 to −0.054. Columns (4)–(6) present the results for subjective uncertainty, which also shows a consistently negative and statistically significant relationship, with coefficients between −0.019 and −0.021. Importantly, Columns (3) and (6) include the average daily EPU over the past 30 days to control for the general level of macroeconomic uncertainty to which households are exposed. A 1% increase in the average daily EPU significantly reduces mean expectations by 5.43% but does not affect subjective uncertainty. Even after accounting for this longer-run component, the coefficient on the daily EPU remains negative and significant. It implies that the initial level of macroeconomic uncertainty does not change households' responses to daily changes in macroeconomic uncertainty. These findings are robust across alternative measures of mean expectations and subjective uncertainty (see Appendix B.3). Taken together, these findings suggest that one-day increases in macroeconomic uncertainty—preceding the survey response—are associated with reductions in both mean expectations and subjective uncertainty among households. While not definitive evidence of causality, the results lend further support to our interpretation by limiting the scope for reverse causality and reinforcing the temporal direction of the relationship.

4.4. The shape of the histogram

This section examines how the shapes of individuals' forecast histograms change in response to macroeconomic uncertainty. In the previous sections, we find that people become increasingly pessimistic during periods of high macroeconomic uncertainty, as evidenced by the negative relationship between macroeconomic uncertainty and mean expectations. However, our results show mixed evidence regarding the relationship between macroeconomic uncertainty and individual subjective uncertainty. Professional forecasters show a positive correlation between macroeconomic and subjective uncertainty, whereas households show the opposite.

To further examine how individuals adjust expectations during periods of increasing macroeconomic uncertainty, we focus on three aspects of their forecast histograms: (1) the lower-bound forecast, defined as the midpoint of the leftmost bin to which the respondent assigns a nonzero probability; (2) the upper-bound forecast, defined similarly for the rightmost bin; and (3) the span, calculated as the difference between the upper- and lower-bound forecasts. The first two measures allow us to assess whether individuals revise the lower or upper ends of their subjective distributions. At the same time, the span reflects the overall range of expectations and serves as an alternative proxy for subjective uncertainty. Table 6 reports the estimates, with robustness checks provided in Appendix B.4.

For professional forecasters (Panel A), the lower bound declines significantly across all macroeconomic uncertainty measures, with coefficients of −0.707 for EPU, −1.656 for the Jurado et al. (2015) index, and −0.550 for forecast disagreement. For the upper bound, only the EPU coefficient is negative and statistically significant. The span increases across all measures, reaching statistical significance for forecast disagreement and the Jurado et al. (2015) index. These results indicate that, under heightened macroeconomic uncertainty, professional forecasters revise down the lower tail of their forecast distribution and, in some cases, expand the overall forecast range. This is consistent with earlier findings of reduced mean expectations and greater subjective uncertainty in such periods.

For households (Panel B), the upper bound declines significantly across all macroeconomic uncertainty measures, with coefficients of −0.159 for EPU, −1.244 for the Jurado et al. (2015) index, and −0.286 for forecast disagreement. The lower-bound coefficient is negative across all measures, but only the forecast disagreement coefficient is statistically significant. The span contracts significantly for EPU and the Jurado et al. (2015) index, while the coefficient of forecast disagreement is negative but not statistically significant. These results indicate that, during periods of elevated macroeconomic uncertainty, households primarily adjust by lowering the upper tail of their forecast distribution and, in some cases, compressing its overall range. This pattern aligns with earlier evidence that heightened macroeconomic uncertainty is associated with lower mean expectations and reduced subjective uncertainty among households.

Fig. 2 illustrates stylized changes in the shape of forecast distributions under heightened macroeconomic uncertainty. Panel (a) shows the case of professional forecasters: when macroeconomic uncertainty increases, the lower-bound forecast declines, shifting the distribution leftward toward more pessimistic expectations and increasing dispersion. Panel (b) depicts the household response: the forecast distribution also shifts leftward but becomes more compressed, with the probability mass concentrating near the pessimistic forecasts.

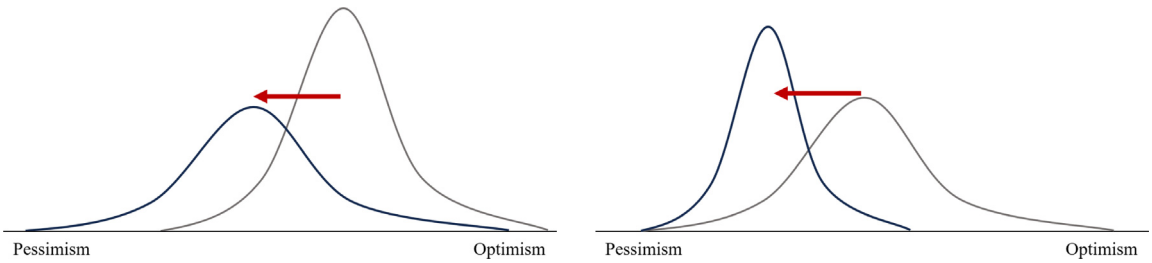
Taken together, the findings deepen our understanding of how macroeconomic uncertainty shapes individual expectations. Professional forecasters respond to heightened uncertainty by lowering the lower bound of their forecast distribution and widening its downside risk. Households, in contrast, respond mainly by truncating the optimistic bound forecast and effectively narrowing the range of their expectations near pessimistic states. This asymmetry highlights that households process and respond to macroeconomic signals differently from professional forecasters.

[8] The Federal Reserve distributes the questionnaire to professional forecasters on the day the US Bureau of Economic Analysis releases the advance GDP estimate. Forecasters typically have a 10-day window to submit their responses.

Table 5
Household's expectations and the Daily Economic Policy Uncertainty index.

	Mean expectation provided by households			Subjective uncertainty (MAM)		
	(1)	(2)	(3)	(4)	(5)	(6)
Log Daily Economic Policy Uncertainty	−0.045***	−0.054***	−0.039***	−0.021**	−0.019**	−0.021***
	(0.017)	(0.016)	(0.015)	(0.0084)	(0.0073)	(0.0076)
Lagged real-time GDP growth	0.0012	0.0029**	0.0021*	−0.00018	0.0010	0.0011*
	(0.00090)	(0.0012)	(0.0012)	(0.00026)	(0.00063)	(0.00065)
Lagged real-time employment rate	0.0013***	0.0021**	0.0014	−0.00042*	−0.000088	0.000020
	(0.00038)	(0.0010)	(0.00090)	(0.00022)	(0.00044)	(0.00049)
Being unemployed	−0.26**	−0.40***	−0.39***	0.060	−0.027	−0.029
	(0.11)	(0.15)	(0.15)	(0.081)	(0.071)	(0.071)
Lagged dependent variable	0.63***	0.064***	0.063***	0.75***	0.073***	0.073***
	(0.0058)	(0.0085)	(0.0085)	(0.0065)	(0.011)	(0.011)
Average daily EPU (past 30 days)			−0.11**			0.018
			(0.044)			(0.018)
Constant	1.13***	4.94***	5.43***	0.47***	3.73***	3.65***
	(0.088)	(0.27)	(0.34)	(0.042)	(0.12)	(0.15)
Number of observations	70 445	68 702	68 702	65 883	64 350	64 350
Number of individuals	11 780	10 037	10 037	10 869	9336	9336
Adjusted R-squared	0.425	0.602	0.602	0.579	0.742	0.742
Year-Month FE	Y	Y	Y	Y	Y	Y
Individual FE	N	Y	Y	N	Y	Y

Note: The standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals (professional forecasters and households). Year-quarter fixed effects for professional forecasters are year and quarter dummies. For households, we use year and month dummies. Individual fixed effects are dummies for individuals. See Appendix B.3 Table 18 for robustness checks.



(a) Professional forecasters (b) Households

Fig. 2. Stylized responses of forecast distributions to heightened macroeconomic uncertainty.

5. Discussion

This section reconciles the empirical findings above with modern decision-making theories under uncertainty. We show that the mechanism that replicates the non-monotonic relationship between macroeconomic and subjective uncertainty also directly implies subjective extrapolation of uncertainty from personal reference (therefore higher in more unstable environments, as in Fermand et al. (2024)) and disproportionately pessimistic average expectations. This mechanism is consistent with standard expected utility theory augmented by either Max-Min or smooth ambiguity preferences. In the following subsections, we first introduce the theoretical concepts of Max-Min expected utility and smooth ambiguity preferences in intuitive, non-technical terms. We then discuss how these frameworks help explain the empirical patterns observed in our data. Lastly, we provide additional and alternative explanations for our empirical findings by examining other literature, including interdisciplinary studies and bounded rationality.

5.1. Uncertainty in macroeconomic models

Following the classic distinction by Knight (1921), uncertainty can be distinguished between risk and ambiguity. Risk refers to situations in which the exact outcomes are uncertain, but their probabilities are known. In contrast, ambiguity refers to situations in which both the outcomes and their associated probabilities are uncertain. Most conventional macroeconomic models conceptualize uncertainty as risk. In this framework, uncertainty shocks are modeled as increases in the standard deviation of exogenous shocks, leading to a rise in the dispersion of agents' expectations (Bloom, 2014; Born et al., 2018; Born & Pfeifer, 2021; Fernández-Villaverde & Guerrón-Quintana, 2020). For risk-averse agents, such increases in dispersion reduce expected utility, consistent with the von Neumann–Morgenstern expected utility theorem. The resulting macroeconomic responses, such as precautionary saving, are thus driven by more pessimistic expectations under risk.

A growing strand of literature extends these models by incorporating ambiguity, recognizing that agents may

Table 6
The shape of forecast histogram of professional forecasters and households.

Panel A: Professional forecasters	Lower-bound forecast			Upper-bound forecast			Span between upper- and lower-bound forecasts		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log Economic Policy Uncertainty	−0.707***			−0.424**			0.301		
	(0.164)			(0.206)			(0.267)		
Log Jurado et al. index		−1.656**			0.561			2.363***	
		(0.690)			(1.324)			(0.815)	
Log forecasters' disagreement			−0.550***			0.0439			0.661**
			(0.202)			(0.283)			(0.331)
Lagged real-time GDP growth	−0.00586	−0.00856	−0.00710	0.0237	0.0273	0.0257	0.0299	0.0368**	0.0337
	(0.00782)	(0.00821)	(0.00825)	(0.0248)	(0.0236)	(0.0239)	(0.0336)	(0.0146)	(0.0325)
Lagged dependent variable	0.547***	0.546***	0.548***	0.536***	0.542***	0.540***	0.577***	0.582***	0.582***
	(0.0487)	(0.0495)	(0.0490)	(0.0602)	(0.0564)	(0.0580)	(0.0707)	(0.0666)	(0.0699)
Constant	3.734***	−0.506	0.201	4.246***	2.528***	2.248***	0.270	2.966***	1.910***
	(0.813)	(0.387)	(0.205)	(0.949)	(0.823)	(0.336)	(1.189)	(0.534)	(0.362)
Number of observations	3761	3761	3761	3761	3761	3761	3761	3761	3761
Number of individuals	146	146	146	146	146	146	146	146	146
Adjusted R-squared	0.68	0.68	0.68	0.80	0.80	0.80	0.80	0.80	0.80
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Survey format FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Panel B: Households	Lower-bound forecast			Upper-bound forecast			Span between upper- and lower-bound forecasts		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log Economic Policy Uncertainty	−0.0172			−0.159***			−0.141**		
	(0.0569)			(0.0481)			(0.0683)		
Log Jurado et al. index		−0.341			−1.244***			−0.893**	
		(0.292)			(0.332)			(0.380)	
Log forecasters' disagreement			−0.214***			−0.286***			−0.0753
			(0.0805)			(0.0755)			(0.0747)
Lagged real-time GDP growth	0.00266	0.00160	0.000240	0.00634***	0.00268	0.00338	0.00360	0.00102	0.00303
	(0.00174)	(0.00191)	(0.00176)	(0.00238)	(0.00217)	(0.00254)	(0.00254)	(0.00265)	(0.00279)
Lagged real-time employment rate	0.00206*	0.00166	0.00116	0.00176	0.000695	0.00109	−0.000395	−0.00105	−0.000180
	(0.00107)	(0.00102)	(0.000914)	(0.00191)	(0.00157)	(0.00154)	(0.00156)	(0.00142)	(0.00162)
Being unemployed	−0.392**	−0.385**	−0.368*	−0.273	−0.250	−0.246	0.119	0.134	0.123
	(0.191)	(0.191)	(0.191)	(0.205)	(0.204)	(0.205)	(0.260)	(0.260)	(0.260)
Lagged dependent variable	0.0542***	0.0542***	0.0540***	0.0639***	0.0637***	0.0636***	0.0914***	0.0913***	0.0914***
	(0.00962)	(0.00962)	(0.00962)	(0.00703)	(0.00704)	(0.00703)	(0.0103)	(0.0103)	(0.0103)
Constant	−3.674***	−3.928***	−4.003***	11.95***	10.55***	10.84***	15.10***	13.96***	14.31***
	(0.434)	(0.357)	(0.333)	(0.439)	(0.397)	(0.377)	(0.582)	(0.480)	(0.479)
Number of observations	69 027	69 027	69 027	69 027	69 027	69 027	69 027	69 027	69 027
Number of individuals	10 057	10 057	10 057	10 057	10 057	10 057	10 057	10 057	10 057
Adjusted R-squared	0.67	0.67	0.67	0.69	0.69	0.69	0.76	0.76	0.76
Year-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: The standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals (professional forecasters and households). Year-quarter fixed effects for professional forecasters are year and quarter dummies. For households, we use year and month dummies. Individual fixed effects are dummies for individuals. See Appendix B.4 Tables 19 and 20 for robustness checks.

not know the true distribution of possible outcomes. In these models, ambiguity is typically represented by a set of prior beliefs rather than a single known distribution, and ambiguity shocks are typically modeled as an increase in the set of prior beliefs over possible scenarios, leading to a lower expectation of the worst-case beliefs relative to the better-case beliefs (Altug et al., 2020; Ilut & Schneider, 2014, 2022; Marinacci, 2015; Piccillo & Poonpakdee, 2023). Two influential frameworks that distinguish between risk and ambiguity are the Max-Min expected utility model of Gilboa and Schmeidler (1989) and the smooth ambiguity preferences model of Klibanoff et al. (2005, 2009). These models extend the standard expected utility framework to allow for uncertainty not only over outcomes but also over the probabilities themselves.

These two models differ in how agents process uncertainty in the face of ambiguity. In the Max-Min expected utility framework, ambiguity-averse agents make decisions based on the worst-case scenario among a set of plausible probability distributions. They act as if the worst-case scenario were true, leading to highly pessimistic behavior in the face of ambiguity (Gilboa & Schmeidler, 1989). In contrast, the smooth ambiguity preferences model allows for more flexible responses: agents consider multiple probability distributions and assign subjective probabilities across them. The degree of ambiguity aversion determines how much weight agents place on pessimistic scenarios. As ambiguity aversion increases, agents assign greater subjective probability to pessimistic scenarios. Notably, the Max-Min expected

utility is a special case of smooth ambiguity preferences. When the degree of ambiguity aversion or the level of ambiguity is sufficiently high, the agent assigns 100% probability to the worst-case scenario (Klibanoff et al., 2005, 2009; Marinacci, 2015).

While both risk-based and ambiguity-based models generate pessimistic expectations, they do so via distinct mechanisms. Under risk, agents respond to increases in volatility by engaging in intertemporal consumption smoothing—postponing consumption to hedge against future risks. In contrast, ambiguity-averse agents in frameworks such as Max-Min expected utility (Gilboa & Schmeidler, 1989) or smooth ambiguity preferences (Klibanoff et al., 2005) seek to smooth expected utilities across plausible scenarios. The resulting first-order conditions imply a lower expected utility. This intratemporal smoothing of expected utility—across scenarios rather than across time—produces first-moment effects that are conceptually analogous to those under risk but arise from fundamentally different preferences.

These theoretical insights motivate our effort to explain the puzzling patterns observed in our empirical results. Under rising macroeconomic uncertainty, both professional forecasters and households revise their mean income expectations downward. However, their subjective uncertainty, as well as the changes in the shape of their forecast histograms, differ markedly. In the next section, we explore how the decision-making theories discussed above can help reconcile these differences and shed light on the underlying mechanisms.

5.2. Reconciling with empirical findings

This section connects the empirical findings to decision-making theories under ambiguity and risk. To illustrate this linkage, we begin by outlining an analytical framework. Ambiguity (A) is defined as an increase in the number of prior beliefs, leading to a lower expectation of the worst-case relative to the best-case scenarios (Ilut & Schneider, 2014, 2022; Piccillo & Poonpakdee, 2023) whereas risk (R) is defined as an increase in the standard deviation of the distribution in the prior beliefs (Bloom, 2014; Born et al., 2018; Born & Pfeifer, 2021; Fernández-Villaverde & Guerrón-Quintana, 2020). Without loss of generality, this implies that:

$$A = E^b(\dots) - E^w(\dots) \tag{1}$$

$$R = \sigma_i \text{ for } i \in \{b, w\} \tag{2}$$

where E^b and E^w are the expectations of future outcomes in the best and worst-case scenarios, respectively. σ_i is the standard deviation of the probability distribution in scenario i.

In this paper, people's expectations are retrieved from surveys of professional forecasters and households. Although expectations are formed under risk and ambiguity, the surveys do not ask for the distributions of different scenarios. Delavande et al. (2022) suggest that population surveys capture the subjective expectations of individuals under both risk and ambiguity. Therefore, we assume that the responses reflect the combined subjective distributions of the best- and worst-case scenarios. Let Y denote the growth rate of income or GDP, and let $P(w)$ be the subjective probability of the worst-case scenario. The respondent's mean expectation $E(Y)$ and subjective uncertainty $\sigma(Y)$ follow

$$E(Y) = P(w)E^w(Y) + (1 - P(w))E^b(Y) \tag{3}$$

$$\sigma(Y) = \Big[P(w)\left(\sigma_w^2(E(Y) - E^w(Y))^2\right) + (1 - P(w))\left(\sigma_b^2(E(Y) - E^b(Y))^2\right)\Big]^{0.5} \tag{4}$$

The equations above describe our variables of interest as studied in the earlier sections. In instances where the respondent displays signs of personal instability, such as lower levels of education or shorter job tenure, the spread between best- and worst-case expectations is often greater than that of an individual with stable circumstances. This is because personal instability can lead to a wider range of potential outcomes for career and personal progression. All else being equal, Eq. (4) suggests that individuals with personal instability are likely to have a higher level of subjective uncertainty. This is consistent with the findings in Fermand et al. (2024).

We now introduce Max-Min expected utility and smooth ambiguity preferences to determine $P(w)$. Under the Max-Min expected utility framework, the respondent is assumed to be extremely ambiguity averse and to assign full weight to the worst-case scenario, implying $P(w) = 1$. In contrast, under the smooth ambiguity preferences, $P(w)$ varies with the degree of ambiguity aversion.

Let $\mathbf{a}$ denote a set of possible outcomes. In our context, we can interpret $\mathbf{a}$ as the possible outcomes of future income growth for households or future output growth for professional forecasters. Given the Bayesian probability of the worst-case scenario μ, the subjective probability of the worst-case scenario $\hat{\mu}$ can be described as the combination of Bayesian belief μ and belief distortion $\xi^w(\mathbf{a})$.

$$\hat{\mu} = \mu\xi^w(\mathbf{a}) \tag{5}$$

$$\xi^w(\mathbf{a}) = \frac{\phi'(E^w(\nu(\mathbf{a})))}{\mu\phi'(E^w(\nu(\mathbf{a}))) + (1 - \mu)\phi'(E^b(\nu(\mathbf{a})))} \tag{6}$$

where ν is a utility function whose concavity depends on risk aversion $\frac{\nu''}{\nu'} < 0$ and ϕ is a smooth ambiguity function with concavity increasing with ambiguity aversion $\frac{\phi''}{\phi'} < 0$.

Assume that people's income expectations align with smooth ambiguity preferences and behave according to Eqs. (5) and (6), the probability of the worst case in the survey expectation $P(w)$ is equivalent to the subjective probability of the worst case $\hat{\mu}$. It follows that an increase in ambiguity, as quantified by Eq. (1) leads to an increase in $\xi^w(\mathbf{a})$ and $P(w)$, implying that the respondents put more weight on the worst-case scenario.[9]

[9] For more detail about smooth ambiguity preferences, see Marinacci (2015) and Klibanoff et al. (2005).

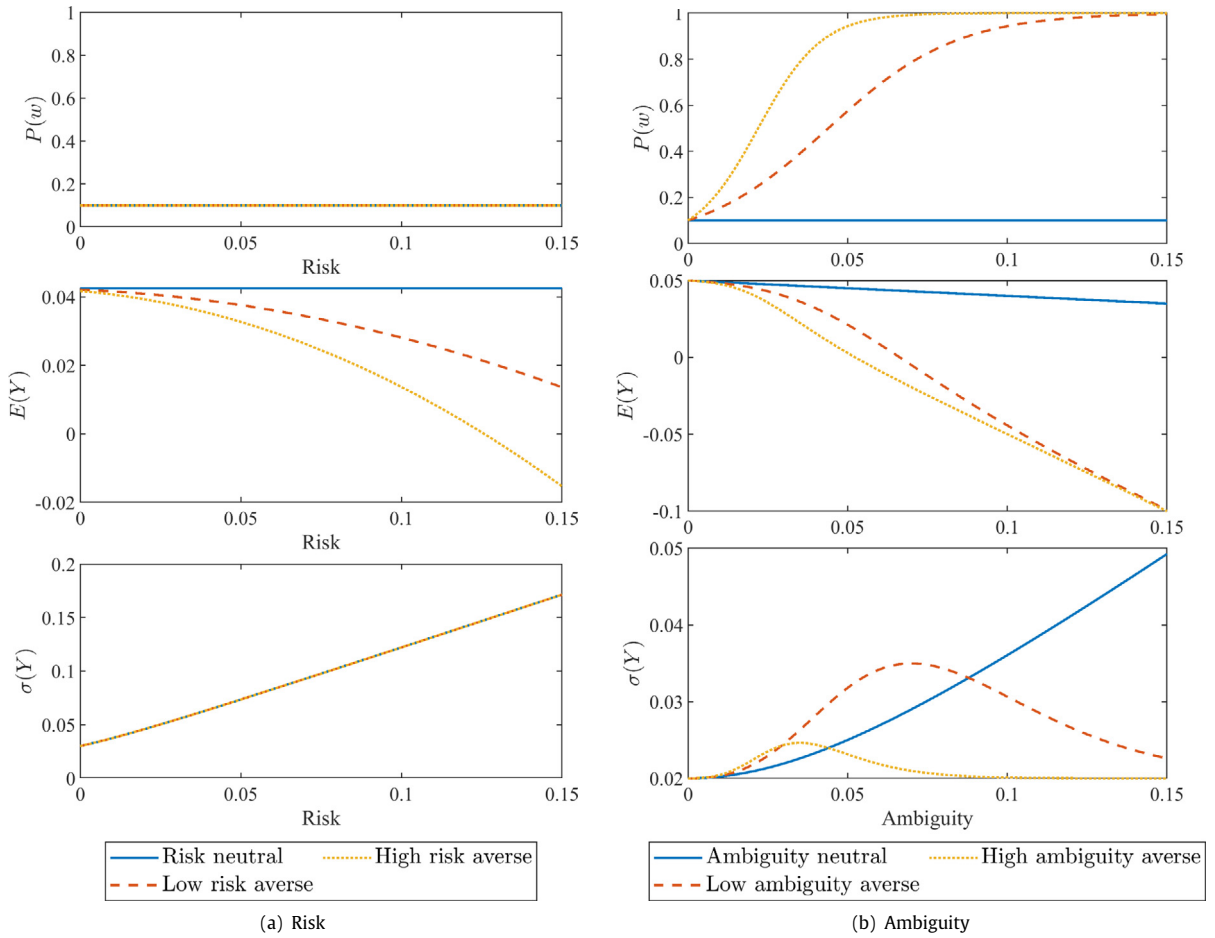


Fig. 3. Stylized effects of uncertainty on subjective probability and expectations.
Note: These figures are generated from the analytical framework explained in Section 5.2. The details of the model are provided in Appendix C.

We now have an analytical framework to discuss our empirical results. Fig. 3 illustrates the stylized relationships between uncertainty and income expectations in this framework. The left column shows the effects of risk, and the right column shows those of ambiguity. The three rows show the subjective probability of the worst-case scenario $P(w)$, the mean $E(Y)$, and the subjective uncertainty $\sigma(Y)$, respectively.

It follows directly from the framework that an increase in ambiguity or risk leads to a lower expected utility, since people are both risk-averse and ambiguity-averse. Since expected utility is positively associated with expected income, this implies that income expectations decline as risk or ambiguity rises—consistent with our empirical finding that mean expectations decline during periods of heightened macroeconomic uncertainty.

The positive relationship between macroeconomic and subjective uncertainty observed among professional forecasters is also consistent with the model. As shown in Table 6 Section 4.4, professional forecasters revise the lower bound of their forecast distribution downward and widen the span, indicating greater dispersion of probabilistic beliefs. This aligns with the model linking an increase in risk or moderate levels of ambiguity to an expansion of subjective uncertainty. In our framework, such behavior arises when the pessimistic belief distortion ξ^w remains small and does not dominate the effect of risk. That is the case when people are ambiguity neutral or when the level of ambiguity is sufficiently low. Under these conditions, people respond to heightened macroeconomic uncertainty by broadening the perceived range of possible outcomes, consistent with the empirical evidence reported for professional forecasters.

More interestingly, the negative relationship between macroeconomic and subjective uncertainty observed in household data arises naturally within this framework as a consequence of ambiguity. As ambiguity increases, ambiguity-averse households exhibit stronger belief distortions ξ^w and assign higher subjective probabilities to the worst-case scenario $P(w)$, reducing the weight on the best-case scenario. This mechanism directly maps onto the empirical findings reported in Table 6 and discussed in Section 4.4. When macroeconomic uncertainty

rises, household forecast histograms shift toward the pessimistic side of the distribution—primarily through a compression of the upper tail—resulting in lower subjective uncertainty. Consistent with the theoretical prediction, the decline in subjective uncertainty under higher ambiguity occurs because optimistic outcomes receive diminishing weight as households' beliefs converge toward the Max–Min benchmark ($P(w) \to 1$). Therefore, the contraction of the upper bound documented in Table 6 represents a behavioral manifestation of ambiguity aversion.

The negative relationship between subjective and macroeconomic uncertainty suggests that pessimistic belief distortions, ξ^w, play a more prominent role in shaping households' expectations. In contrast, professional forecasters—who exhibit a positive relationship between the two—tend to exhibit a lower degree of such distortions or perceive macroeconomic uncertainty more in terms of risk than ambiguity.[10]

5.3. Alternative explanations

An examination of interdisciplinary literature on the role of uncertainty in belief formation provides new insights that can also help explain the puzzle highlighted here.

The theory of rational inattention (Sims, 2003) posits that agents have different capacities to process information and, given the cost of obtaining it, can rationally choose to allocate their attention accordingly. During periods of uncertainty, when the cost of obtaining more information is higher, people may become inattentive to costly information and form expectations based on more familiar, accessible information, thereby reducing the spread of forecast distributions. Empirical research has found that households are rationally inattentive; when cost information decreases, households' forecasts of macroeconomic variables are closer to those of professional forecasters (Carroll, 2003; Lamla & Lein, 2014).

Conviction Narrative theory (Tuckett & Nikolic, 2017) identifies two psychological states under uncertainty: integrated and divided. Individuals in the divided state ignore conflicting information and focus only on evidence that reinforces their prior beliefs. This behavior decreases subjective uncertainty. For example, fund managers need a conviction narrative for each stock to make investment decisions under uncertainty (Chong & Tuckett, 2015). Moreover, Tuckett and Nikolic (2017) show that before the Global Financial Crisis, there was a decrease in the thematic diversity of news and textual data, suggesting that the divided mental state might have been prevalent at the time.

In their experimental study, Piccillo and Van Den Hurk (2020) demonstrate that when ambiguity-averse participants are presented with a salient uncertainty, they tend to detect illusory patterns, such as human faces, in blurry pictures that contain only noise. The results of the experiment suggest that ambiguous attitudes shape people's views of the world, making them more likely to perceive structure and patterns even in the absence of a structured signal. This mechanism also provides an empirical implication for the spread of fake news in times of greater uncertainty, as explored by t'Serstevens et al. (2022).

Finally, mechanical factors might explain why households' subjective uncertainty does not increase with macroeconomic uncertainty. First, Coibion et al. (2024) find that European households consistently revise their subjective uncertainty downward, regardless of whether they receive treatment information about mean forecasts or forecasters' disagreement—one of the macroeconomic uncertainty indices in our analysis. A similar pattern may apply to US households in our sample. Second, in periods of higher macroeconomic uncertainty, households might be exposed to a different level of uncertainty than professional forecasters (who, by definition, have a job). In either case, researchers studying the impact of macroeconomic risk or ambiguity should be aware of these correlations when modeling household behavior, especially in periods of high uncertainty.

6. Conclusion

This paper empirically studies the relationship between macroeconomic uncertainty and individual expectations using the US Survey of Professional Forecasters and the Survey of Consumer Expectations of US households. Our main findings are as follows.

First, mean expectations of income and GDP growth decrease in periods of high macroeconomic uncertainty for both groups. This result aligns with several economic and financial models, which posit that people become more pessimistic as macroeconomic uncertainty increases.

Second, we observe that subjective uncertainty has a non-monotonic relationship with macroeconomic uncertainty. A positive relationship is observed among professional forecasters, whereas no significant positive relationship is found in the household data. In fact, households often exhibit a negative correlation, which challenges the conventional assumption that higher uncertainty directly translates into greater subjective belief dispersion.

Third, professional forecasters and households adjust their forecast distributions asymmetrically in periods of high macroeconomic uncertainty. Professional forecasters revise their lower-bound forecasts downward and widen the overall span, implying greater dispersion of probabilistic beliefs. Households, in contrast, primarily reduce their upper-bound forecasts, compressing the range of possible outcomes. This asymmetry produces similarly pessimistic mean expectations but diverging patterns of subjective uncertainty.

Finally, we reconcile these empirical findings with established theories of decision-making under uncertainty. Specifically, we present an analytical framework that shows that the observed patterns are consistent with

[10] For an intuitive explanation of how households and professional forecasters might differ, Enke and Graeber (2023) shows that the magnitude of biases depends on cognitive uncertainty, which increases in the complexity of decisions. This suggests that households and professional forecasters react differently to macroeconomic uncertainty due to the difference in perceived complexity or cognitive uncertainty.

Max-Min expected utility and smooth ambiguity preferences in which uncertainty is distinguished between risk (uncertainty about outcomes) and ambiguity (uncertainty about their likelihoods). As predicted by this framework, ambiguity-averse households exhibiting pessimistic belief distortions place greater weight on pessimistic outcomes at the expense of optimistic ones when ambiguity rises. This leads to reduced subjective uncertainty during periods of heightened macroeconomic uncertainty.

Our study offers a new insight into how individual expectations behave during periods of high macroeconomic uncertainty. This insight might serve as a new basis for further research. Perhaps most relevantly, current economic models do not yet seem to offer a fully empirically supported mechanism of the impact of uncertainty on economic decision-making through belief channels. We hope that these empirical findings, together with the theoretical insights we offer, can inspire new and empirically supported models of the impact of macroeconomic uncertainty. Furthermore, to clearly assess individual preferences over uncertain outcomes, we used forecasts of income and GDP. New research focusing on expectations of inflation or unemployment rates could complement our approach. Finally, our analytical framework, which closes the gap between survey expectations and decision-making theories, may help stimulate future work on survey data from large population samples.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Data

We study the relationship between aggregate uncertainty and individual expectations using two data sets: the survey of US professional forecasters and the survey of US consumer expectations. In the following sections, we describe the survey questions and the summary statistics for each dataset.

A.1. US household data

The US household data comes from the Survey of Consumer Expectations, an online monthly survey conducted by the Federal Reserve Bank of New York since 2013. The respondents, who are the heads of households, are surveyed for up to 12 consecutive months. Fig. 4 displays the questionnaire where mean expectations were asked. Fig. 5 shows the questionnaire where probabilistic distributions were asked.

We exclude observations in which the expected income growth rate lies in the top or bottom fifth percentile within each year, as well as responses that allocate 10% to every bin. For subjective uncertainty, we further exclude cases with constant reported uncertainty—that is, respondents who assign the same probability to each bin throughout the entire series. Table 7 contains the corresponding summary statistics. The number of observations exceeds 96,000, the cross-section comprises around 15,000 respondents, and the average time series per individual is 6 months. The minimum subjective uncertainty is zero because the individuals taking the survey report only one specific bin on the histogram. Out of the total respondents in the SCE, 33% reported being unemployed. However, our study focuses on individuals who provided information about their income expectations, which are typically provided by those currently employed. This results in a very small percentage (0.94%) of unemployed individuals being included in our study, as shown in Table 7.

A.2. US professional forecasters

We use the forecast of the next calendar year's real GDP. If the survey was conducted in 2010, the next calendar year's real GDP growth was the 2011 growth rate. The real GDP series of US professional forecasters started in 1968Q4, but only from 1992 did the forecasting variable become consistent. We, therefore, employ the time series from 1992 onward. The US SPF provides only a point forecast of the GDP *level*. The GDP *growth* forecast is calculated by dividing next year's GDP level by the forecast for this year's GDP level. Fig. 6 and Fig. 7 display the sample questionnaire from 2014Q1. We use the answers in the red rectangle to calculate the next-calendar-year real GDP growth.

The survey also asks for the probability that next year's real GDP growth falls into the different intervals. From 1992Q1 to 2009Q1, the lowest bin is $(-\infty, -2\%)$, and the highest bin is $(+6\%, \infty,)$. From 2009 Q2to 2020Q1, the lowest bin is $(-\infty, -3\%)$, and the highest bin is $(+6\%, \infty,)$. From 2020Q2 onward, the lowest bin is $(-\infty, -12\%)$, and the highest bin is $(+16\%, \infty,)$. Fig. 5 shows the sample questionnaire from 2014Q1. We use the answers in the red rectangle to calculate subjective uncertainty. Table 8 presents a summary of the statistics of the GDP expectations and subjective uncertainty of the US SPF. The number of observations is around 4900, with a cross-section of 187 respondents, and the average time series for each individual is around 27 quarters.

Fig. 8 depicts the cross-sectional averages of mean expectations and subjective uncertainty that differ between professional forecasters and households. Fig. 8(a) displays the professional forecasters' data, and the vertical dashed lines denote when the interval bins have been expanded in 2009Q2 and 2020Q2. The mean expectation of next-calendar-year GDP growth is highly cyclical, as it has decreased substantially during crises, except during the coronavirus pandemic. During this period, we do not observe a reduction in the 2020 expected GDP growth since the crisis was unexpected in 2019. Interestingly, the mean expectations increased, indicating that the forecasters expected the economy to recover in 2021, which ultimately occurred. Furthermore, the subjective uncertainty of professional forecasters, which hovered around 1%, spiked sharply after the second dashed line. This could be attributed to the increased macroeconomic uncertainty

Q23v2part2

By about what percent do you expect your earnings to have [increased/decreased as in Q23]? Please give your best guess.

Instruction H9.

Twelve months from now, I expect my earnings to have [increased/decreased] by __ %

Fig. 4. Question 23 V2 Part 2.

Q24

Suppose again that, **12 months from now**, you are working in the exact same ["main" if Q11>1] job at the same place you currently work, and working the exact same number of hours. In your view, what would you say is the percent chance that 12 months from now...

Your earnings on this job, before taxes and deductions, will have...

increased by 12% or more (bin 1)	_____	percent chance
increased by 8% to 12% (bin 2)	_____	percent chance
increased by 4% to 8% (bin 3)	_____	percent chance
increased by 2% to 4% (bin 4)	_____	percent chance
increased by 0% to 2% (bin 5)	_____	percent chance
decreased by 0% to 2% (bin 6)	_____	percent chance
decreased by 2% to 4% (bin 7)	_____	percent chance
decreased by 4% to 8% (bin 8)	_____	percent chance
decreased by 8% to 12% (bin 9)	_____	percent chance
decreased by 12% or more (bin 10)	_____	percent chance
Total	**100**	

Fig. 5. Question 24.

Table 7
Summary statistics of US households.

Panel A: Gross personal income growth expectations	Mean	Std Dev.	Min	Max	N
Mean expectation (%)					
provided by respondents	3.43	2.88	−4.9	19.5	96,467
using MAM	3.16	2.77	−14	14	96,408
using GBD	3.15	3.24	−29.68	36.28	96,228
Subjective uncertainty (%)					
using MAM	1.83	1.83	0.00	14	90,154
using GBD	1.68	1.94	0.37	21.61	96,228
using EPRS	41.09	27.48	0.00	89.98	90,154
Histogram shape (%)					
Lower-bound forecast	0.13	4.81	−14	14	96,757
Upper-bound forecast	6.54	4.86	−14	14	96,757
Span between upper- and lower-bound forecasts	6.40	7.61	0.00	28.00	96,757
Panel B: Demographics					
Female	47.41%				
Unemployed	0.94%				
Age	44.99	12.51	17	99	

during the pandemic and the change in survey format. Fig. 8(b) illustrates the households' data. Their mean expectations and subjective uncertainty move together, unlike those of professional forecasters. At the beginning of the pandemic, the average household income expectation decreased to its lowest point at 2.58%, then increased, all while subjective uncertainty increased. However, there

Section 2. Real GDP and Its Components

Chain-weighted (2009$)	L / G	Quarterly Data 2013:Q4	2014:Q1	2014:Q2	2014:Q3	2014:Q4	2015:Q1	Annual Data[a] 2013	2014	2015	2016	2017
12. Real GDP		15965.6						15767.1				
13. Real Personal Cons Expenditures		10832.8						10728.2				
14. Real Nonres Fixed Investment		2013.5						1982.1				
15. Real Res Fixed Investment		486.5						486.0				
16. Real Fed Government C & GI		1125.2						1157.5				
17. Real State & Local Govt C & GI		1745.4						1739.7				
18. Real Change in Private Inventories	L	127.2						85.4				
19. Real Net Exports of Goods & Services	L	-370.1						-409.1				

Fig. 6. Section 2 (2014 Q1 form).

Section 4A. Probabilities of Changes in Real GDP (Y/Y)

Please indicate what probabilities you would attach to the various possible percentage changes (annual-average over annual-average) in chain-weighted real GDP. The probabilities of these alternative forecasts should add up to 100. For the last two years, please provide answers only if you normally forecast out that far.

	Probability of indicated percent change in real (chain-weighted) GDP			
	2013-2014	2014-2015	2015-2016	2016-2017
+6 percent or more				
+5.0 to +5.9 percent				
+4.0 to +4.9 percent				
+3.0 to +3.9 percent				
+2.0 to +2.9 percent				
+1.0 to +1.9 percent				
+0.0 to +0.9 percent				
-1.0 to -0.1 percent				
-2.0 to -1.1 percent				
-3.0 to -2.1 percent				
Decline more than 3 percent				
TOTAL	0	0	0	0

Fig. 7. Section 4A (2014 Q1).

Table 8
Summary statistics of the real GDP growth expectation by SPF.

	Mean	Std Dev.	Min	Max	N
Mean expectation(%)					
provided by respondents	2.64	0.92	−6.60	8.63	4,975
using MAM	2.41	0.91	−2.48	7.80	4626
using GBD	2.42	0.86	−2.50	7.27	4145
Subjective uncertainty (%)					
using MAM	1.07	0.51	4.77e−08	5.33	4626
using GBD	0.87	0.39	0.004	4.02	4145
Histogram shape (%)					
Lower-bound forecast	0.30	2.01	−13.55	5.45	4,626
Upper-bound forecast	5.23	2.34	−0.55	17.45	4,626
Span between upper- and lower-bound forecasts	4.93	3.45	0.00	31.00	4,626

does not appear to be any evident structural shift in the household data.

Appendix B. Robustness checks

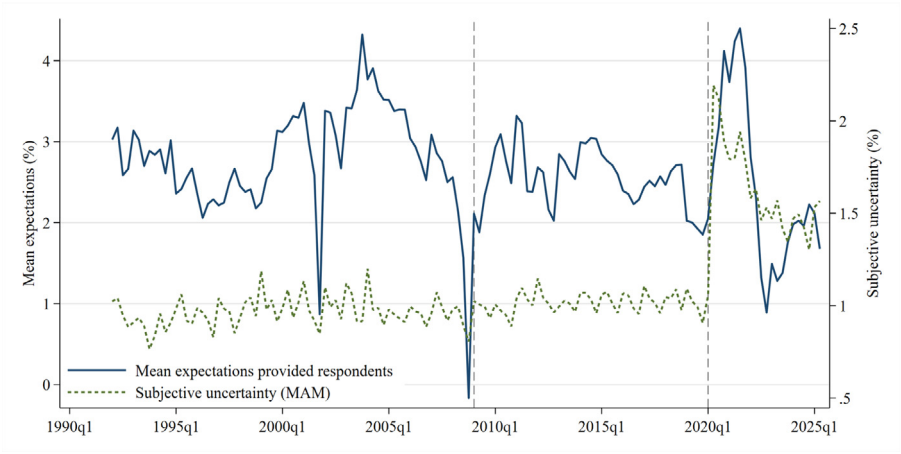
This section reports the full estimates of all regressions.

B.1. Robustness tests for mean expectations

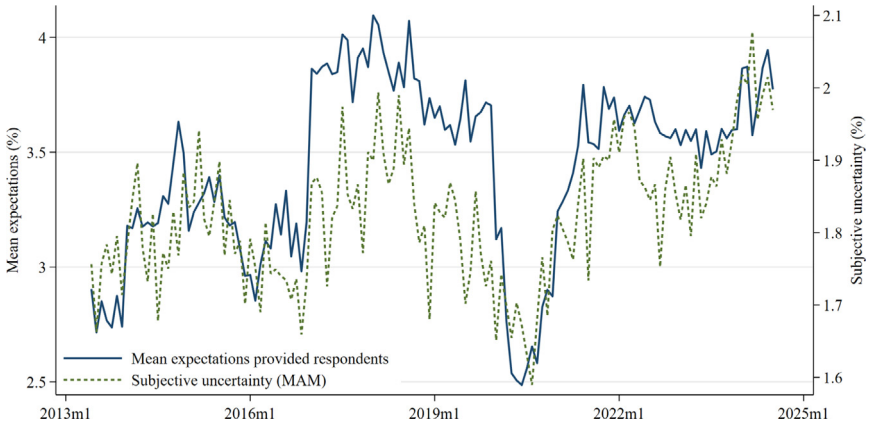
See Tables 9 and 10.

B.2. Robustness tests for subjective uncertainty

See Tables 11 and 12.



(a) Professional forecaters



(b) Households

Fig. 8. Cross-sectional averages of mean expectations provided by the respondents and subjective uncertainty (MAM).

Time frame of the survey of professional forecasters

See Table 13.

Numeracy levels of households

See Table 14.

Personal instability

See Tables 15–17.

B.3. Robustness tests for endogineity

See Table 18.

B.4. Robustness tests for the shape of histogram

See Tables 19 and 20.

Appendix C. Functions to generate Fig. 3

This section describes the functions used to generate the stylized relationship between uncertainty and expectations in Fig. 3. We consider two sources of uncertainty: (1) risk and (2) ambiguity. The model is evaluated under two possible scenarios:

- w: the worst-case scenario
- b: the best-case scenario

The Bayesian prior probability of the worst-case scenario is denoted by $\mu = 0.1$.

Common structure

Let σ_k denote the standard deviation of the outcomes in scenario $k \in \{b, w\}$ and $E^k(Y)$ the corresponding expected outcome. The risk-adjusted expected outcome in scenario k is given by:

$$\nu\left(E^k(Y)\right) = E^k(Y) - 2 \cdot \text{RA} \cdot \sigma_k^2, \tag{7}$$

where $\text{RA} \geq 0$ is the coefficient of risk aversion and the factor 2 is a scaling used in the simulations. Based

Table 9
Professional forecasters' mean expectations.

	Mean expectations directly provided by the respondents									Mean expectations from mass-at-mid point method								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.805***	−0.788***	−0.817***							−0.706***	−0.682***	−0.706***						
	(0.162)	(0.161)	(0.159)							(0.147)	(0.144)	(0.140)						
Log JU				−1.985**	−1.905**	−2.132***							−1.497**	−1.342*	−1.481**			
				(0.776)	(0.827)	(0.805)							(0.689)	(0.723)	(0.698)			
Log Disagreement							−0.435**	−0.396*	−0.463**							−0.428**	−0.377**	−0.420**
							(0.215)	(0.217)	(0.214)							(0.189)	(0.189)	(0.189)
Lagged real-time GDP growth		0.00688*	0.00917**		0.00387	0.00556		0.00719	0.00912		0.00968**	0.0110**		0.00803	0.00903		0.00969*	0.0108*
		(0.00377)	(0.00442)		(0.00473)	(0.00491)		(0.00555)	(0.00601)		(0.00439)	(0.00435)		(0.00558)	(0.00556)		(0.00561)	(0.00576)
Lagged dependent variable	0.561***	0.556***	0.466***	0.547***	0.545***	0.451***	0.557***	0.553***	0.461***	0.534***	0.530***	0.447***	0.525***	0.523***	0.438***	0.532***	0.528***	0.445***
	(0.0463)	(0.0462)	(0.0444)	(0.0475)	(0.0471)	(0.0460)	(0.0473)	(0.0464)	(0.0442)	(0.0340)	(0.0343)	(0.0371)	(0.0351)	(0.0351)	(0.0389)	(0.0336)	(0.0338)	(0.0363)
Survey between 2009Q2-2019Q4	−1.336***	−1.332***	−1.145***	−1.219***	−1.210***	−1.039***	−0.877***	−0.885***	−0.679***	−0.669***	−0.673***	−0.528***	−0.538***	−0.525***	−0.387**	−0.265	−0.284	−0.126
	(0.222)	(0.211)	(0.205)	(0.234)	(0.233)	(0.219)	(0.251)	(0.240)	(0.239)	(0.168)	(0.154)	(0.165)	(0.176)	(0.173)	(0.176)	(0.198)	(0.180)	(0.195)
Survey since 2020Q1	−0.215	−0.178	0.108	−0.170	−0.145	0.134	0.653	0.628	1.017**	0.493**	0.535***	0.780***	0.542**	0.588***	0.829***	1.322***	1.284***	1.593***
	(0.186)	(0.182)	(0.195)	(0.197)	(0.204)	(0.209)	(0.414)	(0.426)	(0.433)	(0.204)	(0.172)	(0.185)	(0.220)	(0.215)	(0.220)	(0.360)	(0.363)	(0.374)
Constant	5.184***	5.102***	5.517***	0.331	0.372	0.542	1.232***	1.242***	1.507***	4.692***	4.568***	4.906***	0.557	0.629	0.793**	1.208***	1.213***	1.433***
	(0.792)	(0.795)	(0.782)	(0.467)	(0.485)	(0.465)	(0.231)	(0.228)	(0.223)	(0.707)	(0.691)	(0.674)	(0.402)	(0.417)	(0.398)	(0.182)	(0.181)	(0.174)
Number of observations	4040	4040	4025	4040	4040	4025	4040	4040	4025	3774	3774	3761	3774	3774	3761	3774	3774	3761
Number of individuals	165	165	150	165	165	150	165	165	150	159	159	146	159	159	146	159	159	146
Adjusted R-squared	0.62	0.62	0.63	0.61	0.61	0.63	0.60	0.60	0.62	0.60	0.60	0.61	0.59	0.59	0.60	0.59	0.59	0.60
Quarter-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y
	Mean expectations from generalized beta distribution method																	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)									
Log EPU	−0.616***	−0.592***	−0.608***															
	(0.129)	(0.126)	(0.124)															
Log JU				−1.081	−0.910	−1.054												
				(0.668)	(0.705)	(0.675)												
Log Disagreement							−0.389**	−0.341**	−0.380**									
							(0.168)	(0.170)	(0.170)									
Lagged real-time GDP growth		0.00878*	0.00966**		0.00836	0.00883		0.00880	0.00944									
		(0.00487)	(0.00469)		(0.00579)	(0.00574)		(0.00588)	(0.00586)									
Lagged dependent variable	0.515***	0.511***	0.432***	0.510***	0.509***	0.427***	0.514***	0.512***	0.432***									
	(0.0323)	(0.0326)	(0.0332)	(0.0324)	(0.0325)	(0.0339)	(0.0316)	(0.0319)	(0.0318)									
Survey between 2009Q2-2019Q4	−0.535***	−0.540***	−0.417**	−0.385*	−0.373*	−0.263	−0.186	−0.205	−0.0726									
	(0.186)	(0.175)	(0.172)	(0.209)	(0.206)	(0.197)	(0.208)	(0.194)	(0.193)									
Survey since 2020Q1	0.612**	0.642**	0.870***	0.674**	0.714**	0.933***	1.350***	1.304***	1.590***									
	(0.261)	(0.246)	(0.244)	(0.283)	(0.278)	(0.276)	(0.364)	(0.376)	(0.377)									
Constant	4.307***	4.184***	4.482***	0.816**	0.894**	1.056***	1.259***	1.262***	1.483***									
	(0.622)	(0.608)	(0.596)	(0.391)	(0.408)	(0.389)	(0.172)	(0.171)	(0.171)									
Number of observations	3307	3307	3292	3307	3307	3292	3307	3307	3292									
Number of individuals	150	150	135	150	150	135	150	150	135									
Adjusted R-squared	0.60	0.60	0.61	0.59	0.59	0.60	0.59	0.59	0.60									
Quarter-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y									
Individual FE	N	N	Y	N	N	Y	N	N	Y									

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 10
Households' mean expectations.

	Mean expectations directly provided by the respondents									Mean expectations from mass-at-mid point method								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.101***	−0.0931***	−0.0982**							−0.0552*	−0.0467	−0.0626						
	(0.0298)	(0.0289)	(0.0378)							(0.0313)	(0.0306)	(0.0399)						
Log JU				−0.196	−0.0992	−0.549**							−0.414***	−0.356*	−0.601**			
				(0.152)	(0.164)	(0.222)							(0.154)	(0.181)	(0.239)			
Log Disagreement							−0.121***	−0.112**	−0.176***							−0.110**	−0.0957*	−0.196***
							(0.0421)	(0.0443)	(0.0566)							(0.0448)	(0.0490)	(0.0640)
Lagged real-time GDP growth		0.00124	0.00300**		0.00109	0.00144		0.000364	0.00118		0.00169***	0.00490***		0.000473	0.00310**		0.000862	0.00278*
		(0.000810)	(0.00118)		(0.000982)	(0.00123)		(0.00101)	(0.00123)		(0.000455)	(0.00139)		(0.000697)	(0.00132)		(0.000742)	(0.00143)
Lagged real-time employment rate		0.00115***	0.00195**		0.00134***	0.00159*		0.00116***	0.00154*		0.00110***	0.00130		0.000707	0.000729		0.000979**	0.000655
		(0.000342)	(0.000983)		(0.000491)	(0.000889)		(0.000327)	(0.000792)		(0.000380)	(0.000973)		(0.000443)	(0.000808)		(0.000388)	(0.000713)
Being unemployed		−0.262**	−0.407***		−0.262**	−0.398***		−0.256**	−0.391***		−0.361***	−0.313**		−0.356***	−0.302**		−0.355***	−0.294**
		(0.111)	(0.147)		(0.110)	(0.147)		(0.110)	(0.147)		(0.0925)	(0.141)		(0.0927)	(0.141)		(0.0922)	(0.141)
Lagged dependent variable	0.632***	0.632***	0.0638***	0.632***	0.632***	0.0636***	0.632***	0.632***	0.0636***	0.603***	0.603***	0.0372***	0.603***	0.603***	0.0370***	0.603***	0.603***	0.0368***
	(0.00585)	(0.00585)	(0.00852)	(0.00586)	(0.00585)	(0.00853)	(0.00586)	(0.00585)	(0.00852)	(0.00646)	(0.00646)	(0.00888)	(0.00646)	(0.00646)	(0.00888)	(0.00647)	(0.00646)	(0.00887)
Constant	1.419***	1.379***	5.178***	0.801***	0.857***	4.417***	0.812***	0.821***	4.486***	1.254***	1.210***	4.209***	0.733***	0.769***	3.601***	0.884***	0.897***	3.675***
	(0.163)	(0.160)	(0.331)	(0.0963)	(0.100)	(0.252)	(0.0630)	(0.0633)	(0.249)	(0.154)	(0.150)	(0.333)	(0.0951)	(0.110)	(0.261)	(0.0582)	(0.0596)	(0.235)
Number of observations	70 445	70 445	68 702	70 445	70 445	68 702	70 445	70 445	68 702	70 387	70 387	68 652	70 387	70 387	68 652	70 387	70 387	68 652
Number of individuals	11780	11780	10037	11780	11780	10037	11780	11780	10037	11766	11766	10031	11766	11766	10031	11766	11766	10031
Adjusted R-squared	0.43	0.43	0.60	0.43	0.43	0.60	0.43	0.43	0.60	0.38	0.38	0.58	0.38	0.38	0.58	0.38	0.38	0.58
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y
	Mean expectations from generalized beta distribution method																	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)									
Log EPU	−0.0411	−0.0324	−0.0546															
	(0.0376)	(0.0370)	(0.0452)															
Log JU				−0.445**	−0.371*	−0.538*												
				(0.181)	(0.219)	(0.277)												
Log Disagreement							−0.153***	−0.140***	−0.204***									
							(0.0421)	(0.0474)	(0.0644)									
Lagged real-time GDP growth		0.00226***	0.00590***		0.000946	0.00428***		0.000961	0.00367**									
		(0.000539)	(0.00149)		(0.000845)	(0.00148)		(0.000766)	(0.00155)									
Lagged real-time employment rate		0.00108**	0.00141		0.000609	0.000898		0.000768	0.000699									
		(0.000544)	(0.00109)		(0.000628)	(0.000988)		(0.000473)	(0.000802)									
Being unemployed		−0.278*	−0.258		−0.272*	−0.248		−0.269*	−0.237									
		(0.152)	(0.159)		(0.152)	(0.160)		(0.152)	(0.160)									
Lagged dependent variable	0.532***	0.532***	0.00473	0.532***	0.532***	0.00466	0.532***	0.532***	0.00446									
	(0.0100)	(0.0100)	(0.0131)	(0.0100)	(0.0100)	(0.0131)	(0.0100)	(0.0100)	(0.0131)									
Constant	1.333***	1.287***	4.436***	0.864***	0.908***	3.897***	0.996***	1.007***	3.932***									
	(0.192)	(0.189)	(0.378)	(0.110)	(0.131)	(0.291)	(0.0605)	(0.0626)	(0.268)									
Number of observations	70 183	70 183	68 451	70 183	70 183	68 451	70 183	70 183	68 451									
Number of individuals	11749	11749	10017	11749	11749	10017	11749	11749	10017									
Adjusted R-squared	0.30	0.30	0.52	0.30	0.30	0.52	0.30	0.30	0.52									
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y									
Individual FE	N	N	Y	N	N	T	N	N	Y									

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 11
Professional forecasters' subjective uncertainty.

	Subjective uncertainty from mass-at-mid point method									Subjective uncertainty from generalized beta distribution method								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	0.101**	0.108**	0.0936**							0.0383	0.0383	0.0307						
	(0.0496)	(0.0530)	(0.0379)							(0.0350)	(0.0363)	(0.0310)						
Log JU				0.408	0.481	0.327							0.0940	0.0967	0.0149			
				(0.269)	(0.295)	(0.198)							(0.171)	(0.172)	(0.134)			
Log Disagreement							0.180***	0.200***	0.132***							0.177***	0.183***	0.134***
							(0.0679)	(0.0698)	(0.0451)							(0.0473)	(0.0469)	(0.0365)
Lagged real-time GDP growth		0.00249	0.00213		0.00370	0.00286		0.00356	0.00274		−0.0000245	0.000460		0.000133	0.000376		0.00120	0.00134
		(0.00673)	(0.00414)		(0.00634)	(0.00391)		(0.00636)	(0.00396)		(0.00321)	(0.00194)		(0.00316)	(0.00195)		(0.00278)	(0.00164)
Lagged dependent variable	0.769***	0.769***	0.457***	0.770***	0.771***	0.460***	0.771***	0.772***	0.462***	0.671***	0.671***	0.390***	0.671***	0.671***	0.390***	0.676***	0.676***	0.397***
	(0.0302)	(0.0298)	(0.0529)	(0.0290)	(0.0281)	(0.0514)	(0.0295)	(0.0285)	(0.0519)	(0.0342)	(0.0341)	(0.0449)	(0.0338)	(0.0337)	(0.0448)	(0.0331)	(0.0332)	(0.0434)
Survey between 2009Q2-2019Q4	−0.0339	−0.0368	−0.00512	−0.0139	−0.00967	0.00131	−0.0923***	−0.102***	−0.0604***	−0.0690**	−0.0690**	−0.0421	−0.0730	−0.0728	−0.0563	−0.0925***	−0.0958***	−0.0639**
	(0.0350)	(0.0351)	(0.0256)	(0.0562)	(0.0582)	(0.0391)	(0.0122)	(0.0210)	(0.0166)	(0.0330)	(0.0338)	(0.0338)	(0.0451)	(0.0443)	(0.0398)	(0.0213)	(0.0256)	(0.0276)
Survey since 2020Q1	0.412	0.420	0.691***	0.434	0.452	0.703***	0.127	0.108	0.475**	0.223	0.223*	0.379***	0.223	0.224	0.371***	−0.0356	−0.0428	0.181*
	(0.293)	(0.287)	(0.207)	(0.296)	(0.281)	(0.203)	(0.219)	(0.233)	(0.182)	(0.136)	(0.133)	(0.117)	(0.146)	(0.142)	(0.127)	(0.0952)	(0.106)	(0.103)
Constant	−0.184	−0.221	0.0774	0.519***	0.550***	0.695***	0.359***	0.357***	0.562***	0.118	0.118	0.316***	0.350***	0.351***	0.469***	0.360***	0.360***	0.504***
	(0.214)	(0.238)	(0.184)	(0.164)	(0.176)	(0.132)	(0.0548)	(0.0574)	(0.0669)	(0.160)	(0.169)	(0.152)	(0.106)	(0.105)	(0.0859)	(0.0367)	(0.0375)	(0.0421)
Number of observations	3774	3774	3761	3774	3774	3761	3774	3774	3761	3307	3307	3292	3307	3307	3292	3307	3307	3292
Number of individuals	159	159	146	159	159	146	159	159	146	150	150	135	150	150	135	150	150	135
Adjusted R-squared	0.70	0.70	0.75	0.70	0.70	0.75	0.70	0.71	0.76	0.57	0.57	0.63	0.57	0.57	0.63	0.57	0.57	0.64
Quarter-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 12
Households' subjective uncertainty.

	Subjective uncertainty from mass-at-mid point method									Subjective uncertainty from generalized beta distribution method								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.0401**	−0.0427**	−0.0351**							−0.0141	−0.0154	−0.0163						
	(0.0175)	(0.0176)	(0.0174)							(0.0143)	(0.0147)	(0.0157)						
Log JU				−0.0734	−0.111	−0.194**							0.00446	−0.0231	−0.0957			
				(0.0590)	(0.0675)	(0.0878)							(0.0503)	(0.0639)	(0.0878)			
Log Disagreement							−0.0147	−0.0185	−0.0104							−0.0214*	−0.0284**	0.0143
							(0.0177)	(0.0197)	(0.0187)							(0.0116)	(0.0128)	(0.0209)
Lagged real-time GDP growth		−0.000164	0.00104*		−0.000479	0.000485		−0.000247	0.000995		−0.000368	0.00125**		−0.000418	0.000975		−0.000612*	0.00145**
		(0.000232)	(0.000606)		(0.000359)	(0.000619)		(0.000368)	(0.000685)		(0.000279)	(0.000585)		(0.000405)	(0.000613)		(0.000310)	(0.000670)
Lagged real-time employment rate		−0.000485**	−0.000128		−0.000502*	−0.000255		−0.000382	−0.0000373		−0.0000991	0.000129		−0.0000780	0.0000625		−0.000129	0.000258
		(0.000220)	(0.000420)		(0.000271)	(0.000380)		(0.000235)	(0.000468)		(0.000284)	(0.000356)		(0.000326)	(0.000350)		(0.000251)	(0.000432)
Being unemployed		0.0585	−0.0284		0.0599	−0.0247		0.0593	−0.0283		0.104	−0.0136		0.104	−0.0119		0.105	−0.0156
		(0.0808)	(0.0710)		(0.0899)	(0.0709)		(0.0808)	(0.0709)		(0.0849)	(0.0773)		(0.0851)	(0.0773)		(0.0848)	(0.0771)
Lagged dependent variable	0.749***	0.749***	0.0732***	0.749***	0.749***	0.0731***	0.749***	0.749***	0.0732***	0.720***	0.720***	0.0464***	0.720***	0.720***	0.0464***	0.720***	0.720***	0.0464***
	(0.00650)	(0.00650)	(0.0107)	(0.00651)	(0.00651)	(0.0107)	(0.00650)	(0.00650)	(0.0107)	(0.0118)	(0.0118)	(0.0146)	(0.0118)	(0.0118)	(0.0146)	(0.0118)	(0.0118)	(0.0146)
Constant	0.568***	0.581***	3.814***	0.324***	0.303***	3.542***	0.355***	0.352***	3.625***	0.418***	0.425***	3.110***	0.350***	0.334***	2.981***	0.329***	0.324***	3.044***
	(0.0928)	(0.0934)	(0.149)	(0.0384)	(0.0414)	(0.120)	(0.0262)	(0.0270)	(0.117)	(0.0729)	(0.0748)	(0.143)	(0.0393)	(0.0447)	(0.127)	(0.0296)	(0.0298)	(0.124)
Number of observations	65 883	65 883	64 350	65 883	65 883	64 350	65 883	65 883	64 350	70 183	70 183	68 451	70 183	70 183	68 451	70 183	70 183	68 451
Number of individuals	10 869	10 869	9336	10 869	10 869	9336	10 869	10 869	9336	11 749	11 749	10 017	11 749	11 749	10 017	11 749	11 749	10 017
Adjusted R-squared	0.58	0.58	0.74	0.58	0.58	0.74	0.58	0.58	0.74	0.54	0.54	0.72	0.54	0.54	0.72	0.54	0.54	0.72
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

	Expected probability rank score								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log EPU	−0.734**	−0.775**	−0.680*						
	(0.341)	(0.348)	(0.382)						
Log JU				0.0288	−0.693	−0.617			
				(0.967)	(1.046)	(1.705)			
Log Disagreement							0.324	0.250	0.0518
							(0.236)	(0.279)	(0.397)
Lagged real-time GDP growth		−0.0130	0.000128		−0.0138	−0.000429		−0.00877	0.00217
		(0.00980)	(0.0115)		(0.0108)	(0.0128)		(0.0116)	(0.0131)
Lagged real-time employment rate		−0.00585	−0.00597		−0.00405	−0.00413		−0.00217	−0.00305
		(0.00385)	(0.00553)		(0.00503)	(0.00594)		(0.00458)	(0.00670)
Being unemployed		−0.429	−0.537		−0.428	−0.543		−0.460	−0.565
		(1.248)	(1.245)		(1.249)	(1.245)		(1.246)	(1.241)
Lagged dependent variable	0.664***	0.664***	0.0431***	0.664***	0.664***	0.0431***	0.664***	0.664***	0.0431***
	(0.00454)	(0.00454)	(0.00702)	(0.00454)	(0.00454)	(0.00703)	(0.00454)	(0.00454)	(0.00703)
Constant	17.05***	17.28***	69.03***	13.41***	13.02***	65.31***	13.68***	13.64***	65.67***
	(1.821)	(1.847)	(3.065)	(0.763)	(0.771)	(2.258)	(0.610)	(0.619)	(2.236)
Number of observations	65 883	65 883	64 350	65 883	65 883	64 350	65 883	65 883	64 350
Number of individuals	10 869	10 869	9336	10 869	10 869	9336	10 869	10 869	9336
Adjusted R-squared	0.44	0.44	0.63	0.44	0.44	0.63	0.44	0.44	0.63
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 13
Professional forecasters' subjective uncertainty from 2013Q1.

	Subjective uncertainty from mass-at-mid point method									Subjective uncertainty from generalized beta distribution method								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	0.196*	0.210*	0.193***							0.115	0.117	0.118**						
	(0.104)	(0.107)	(0.0710)							(0.0704)	(0.0700)	(0.0550)						
Log JU				0.870	1.178*	0.811*							0.398	0.487	0.298			
				(0.551)	(0.640)	(0.441)							(0.353)	(0.383)	(0.301)			
Log Disagreement							0.374**	0.483***	0.317***							0.353***	0.411***	0.321***
							(0.147)	(0.166)	(0.108)							(0.0880)	(0.0905)	(0.0616)
Lagged real-time GDP growth		0.00331	0.00274		0.00661	0.00495		0.00676	0.00490		0.000645	0.000991		0.00194	0.00168		0.00373	0.00331**
		(0.00643)	(0.00392)		(0.00537)	(0.00329)		(0.00654)	(0.00348)		(0.00325)	(0.00203)		(0.00283)	(0.00181)		(0.00238)	(0.00141)
Lagged dependent variables	0.731***	0.732***	0.441***	0.737***	0.742***	0.455***	0.742***	0.748***	0.461***	0.669***	0.668***	0.391***	0.671***	0.672***	0.395***	0.682***	0.684***	0.412***
	(0.0373)	(0.0363)	(0.0671)	(0.0324)	(0.0277)	(0.0608)	(0.0330)	(0.0282)	(0.0615)	(0.0462)	(0.0462)	(0.0667)	(0.0442)	(0.0440)	(0.0655)	(0.0424)	(0.0334)	(0.0621)
Survey since 2020Q1	0.401	0.413*	0.644***	0.391*	0.398**	0.641***	−0.0513	−0.168	0.271	0.207*	0.209*	0.346***	0.211**	0.213**	0.360***	−0.233***	−0.295***	−0.0453
	(0.244)	(0.240)	(0.187)	(0.197)	(0.179)	(0.151)	(0.124)	(0.202)	(0.182)	(0.113)	(0.111)	(0.107)	(0.0864)	(0.0815)	(0.0918)	(0.0255)	(0.103)	(0.0565)
Constant	−0.660	−0.733	−0.345	0.809**	0.969**	1.058***	0.585***	0.648***	0.823***	−0.278	−0.292	−0.0435	0.517**	0.564**	0.708***	0.546***	0.581***	0.762***
	(0.491)	(0.508)	(0.342)	(0.332)	(0.378)	(0.267)	(0.132)	(0.145)	(0.111)	(0.338)	(0.337)	(0.281)	(0.217)	(0.232)	(0.177)	(0.0914)	(0.0833)	(0.0697)
Number of observations	1415	1415	1410	1415	1415	1410	1415	1415	1410	1290	1290	1286	1290	1290	1286	1290	1290	1286
Number of individuals	67	67	62	67	67	62	67	67	62	65	65	61	65	65	61	65	65	61
Adjusted R-squared	0.70	0.70	0.75	0.70	0.70	0.75	0.70	0.70	0.75	0.57	0.57	0.63	0.57	0.57	0.63	0.58	0.58	0.64
Quarter-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 14
Households subjective uncertainty by the levels of numeracy.

Panel A: mass-at-mid point	High numeracy									Low numeracy								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.0420**	−0.0434**	−0.0348**							−0.0351	−0.0418	−0.0452						
	(0.0168)	(0.0172)	(0.0155)							(0.0437)	(0.0430)	(0.0494)						
Log JU				−0.120**	−0.175**	−0.203**							0.0779	0.105	−0.206			
				(0.0573)	(0.0672)	(0.0791)							(0.157)	(0.164)	(0.276)			
Log Disagreement							−0.00526	−0.00965	−0.0145							−0.0478	−0.0487	0.0112
							(0.0154)	(0.0181)	(0.0181)							(0.0416)	(0.0452)	(0.0588)
Lagged real-time GDP growth		−0.000434	0.000396		−0.000991**	−0.000191		−0.000427	0.000303		0.000713	0.00343**		0.00119	0.00286		0.000326	0.00364*
		(0.000338)	(0.000536)		(0.000470)	(0.000557)		(0.000466)	(0.000592)		(0.000937)	(0.00169)		(0.000971)	(0.00183)		(0.00113)	(0.00192)
Lagged real-time employment rate		−0.000148	−0.000248		−0.000266	−0.000390		−0.0000129	−0.000175		−0.00173**	0.000263		−0.00142	0.000163		−0.00172**	0.000480
		(0.000234)	(0.000372)		(0.000303)	(0.000332)		(0.000250)	(0.000403)		(0.000816)	(0.000930)		(0.000886)	(0.000973)		(0.000855)	(0.000107)
Being unemployed		0.0576	−0.0305		0.0606	−0.0266		0.0578	−0.0295		0.0556	−0.0476		0.0534	−0.0439		0.0581	−0.0491
		(0.0887)	(0.0709)		(0.0886)	(0.0710)		(0.0886)	(0.0706)		(0.147)	(0.152)		(0.147)	(0.152)		(0.147)	(0.152)
Lagged dependent variable	0.742***	0.741***	0.0678***	0.742***	0.741***	0.0677***	0.742***	0.742***	0.0678***	0.762***	0.762***	0.0756***	0.762***	0.762***	0.0756***	0.762***	0.762***	0.0756***
	(0.00679)	(0.00677)	(0.0106)	(0.00679)	(0.00678)	(0.0106)	(0.00679)	(0.00678)	(0.0106)	(0.0127)	(0.0128)	(0.0194)	(0.0127)	(0.0128)	(0.0193)	(0.0127)	(0.0128)	(0.0193)
Constant	0.576***	0.584***	3.345***	0.296***	0.264***	3.070***	0.362***	0.359***	3.155***	0.564**	0.596***	5.733***	0.436***	0.449***	5.405***	0.348***	0.346***	5.514***
	(0.0898)	(0.0911)	(0.149)	(0.0401)	(0.0434)	(0.131)	(0.0295)	(0.0304)	(0.129)	(0.224)	(0.220)	(0.437)	(0.117)	(0.124)	(0.350)	(0.0735)	(0.0754)	(0.354)
Number of observations	51 255	51 255	50 167	51 255	51 255	50 167	51 255	51 255	50 167	14 616	14 616	14 175	14 616	14 616	14 175	14 616	14 616	14 175
Number of individuals	8209	8209	7121	8209	8209	7121	8209	8209	7121	2653	2653	2212	2653	2653	2212	2653	2653	2212
Adjusted R-squared	0.56	0.56	0.73	0.56	0.56	0.73	0.56	0.56	0.73	0.61	0.60	0.76	0.61	0.60	0.76	0.61	0.60	0.76
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Panel B: generalized beta distribution	High numeracy									Low numeracy								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.0252	−0.0245	−0.0162							0.0159	0.00817	−0.0237						
	(0.0154)	(0.0156)	(0.0149)							(0.0517)	(0.0528)	(0.0493)						
Log JU				−0.0849*	−0.109*	−0.121*							0.294	0.250	−0.0493			
				(0.0472)	(0.0580)	(0.0675)							(0.182)	(0.202)	(0.319)			
Log Disagreement							−0.00851	−0.0104	0.0199							−0.0686	−0.0927*	−0.00505
							(0.0113)	(0.0138)	(0.0191)							(0.0502)	(0.0520)	(0.0669)
Lagged real-time GDP growth		−0.000120	0.000514		−0.000470	0.000159		−0.000165	0.000782		−0.00125	0.00377*		−0.000349	0.00366*		−0.00218	0.00376*
		(0.000280)	(0.000459)		(0.000372)	(0.000482)		(0.000362)	(0.000523)		(0.00166)	(0.00202)		(0.00172)	(0.00216)		(0.00171)	(0.00220)
Lagged real-time employment rate		0.000235	0.000124		0.000153	0.0000231		0.000296	0.000281		−0.00133	0.0000820		−0.000975	0.000105		−0.00163	0.000147
		(0.000246)	(0.000273)		(0.000281)	(0.000266)		(0.000248)	(0.000322)		(0.00117)	(0.00120)		(0.00126)	(0.00129)		(0.00103)	(0.00127)
Being unemployed		0.0956	−0.0352		0.0973	−0.0330		0.0960	−0.0387		0.0972	0.0168		0.0940	0.0175		0.103	0.0168
		(0.110)	(0.0662)		(0.110)	(0.0663)		(0.111)	(0.0657)		(0.137)	(0.173)		(0.137)	(0.172)		(0.137)	(0.172)
Lagged dependent variable	0.681***	0.681***	0.0302**	0.681***	0.681***	0.0302**	0.681***	0.681***	0.0302**	0.758***	0.758***	0.0649**	0.758***	0.758***	0.0649**	0.758***	0.758***	0.0649**
	(0.0147)	(0.0146)	(0.0128)	(0.0147)	(0.0146)	(0.0128)	(0.0147)	(0.0146)	(0.0128)	(0.0176)	(0.0177)	(0.0267)	(0.0176)	(0.0176)	(0.0267)	(0.0176)	(0.0176)	(0.0267)
Constant	0.495***	0.491***	2.646***	0.319***	0.306***	2.505***	0.362***	0.361***	2.587***	0.314	0.355	4.931***	0.566***	0.541***	4.787***	0.333***	0.317***	4.804***
	(0.0806)	(0.0814)	(0.148)	(0.0362)	(0.0409)	(0.130)	(0.0285)	(0.0295)	(0.127)	(0.266)	(0.271)	(0.452)	(0.128)	(0.137)	(0.365)	(0.0802)	(0.0797)	(0.373)
Number of observations	54 114	54 114	52 896	54 114	54 114	52 896	54 114	54 114	52 896	16 057	16 057	15 547	16 057	16 057	15 547	16 057	16 057	15 547
Number of individuals	8780	8780	7562	8780	8780	7562	8780	8780	7562	2962	2962	2452	2962	2962	2452	2962	2962	2452
Adjusted R-squared	0.48	0.48	0.67	0.48	0.48	0.67	0.48	0.48	0.67	0.60	0.60	0.76	0.60	0.60	0.76	0.60	0.60	0.76
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 15
Summary statistics of subjective uncertainty by personal instability.

MAM subjective uncertainty	Mean	S.D.
Job tenure		
Work in main job < 5 years	1.92	1.82
Work in main job ≥ 5 years	1.71	1.82
Education level		
No college degree	1.88	2.20
Have a college degree	1.79	1.60
GBD subjective uncertainty	**Mean**	**S.D.**
Job tenure		
Work in main job < 5 years	1.77	1.93
Work in main job ≥ 5 years	1.57	1.89
Education level		
No college degree	1.82	2.46
Have a college degree	1.59	1.55

Table 16
Households subjective uncertainty (MAM) by personal instability.

Panel A: Job tenure	Main job < 5 Years									Main job ≥ 5 Years								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.0323	−0.0354	−0.0368							−0.0534**	−0.0527**	−0.0316						
	(0.0212)	(0.0216)	(0.0235)							(0.0251)	(0.0253)	(0.0247)						
Log JU				−0.189***	−0.185**	−0.277*							0.0140	−0.0265	−0.0923			
				(0.0705)	(0.0829)	(0.140)							(0.104)	(0.102)	(0.105)			
Log Disagreement							−0.0315	−0.0252	0.00469							−0.00637	−0.0157	−0.0186
							(0.0226)	(0.0243)	(0.0294)							(0.0203)	(0.0230)	(0.0300)
Lagged real-time GDP growth		0.00130***	0.00202**		0.000678	0.00119		0.00113**	0.00215**		−0.00112*	0.000200		−0.00109*	−0.0000311		−0.00115	0.0000516
		(0.000399)	(0.000871)		(0.000551)	(0.000899)		(0.000504)	(0.00101)		(0.000608)	(0.000789)		(0.000644)	(0.000811)		(0.000757)	(0.000885)
Lagged real-time employment rate		−0.000890**	−0.000507		−0.00105**	−0.000736*		−0.000837**	−0.000340		0.000503	0.000525		0.000656	0.000520		0.000651*	0.000562
		(0.000384)	(0.000481)		(0.000437)	(0.000430)		(0.000415)	(0.000577)		(0.000341)	(0.000510)		(0.000396)	(0.000524)		(0.000352)	(0.000548)
Being unemployed		0.0439	−0.101		0.0470	−0.0945		0.0460	−0.101		0.0600	−0.0145		0.0600	−0.0135		0.0604	−0.0144
		(0.103)	(0.115)		(0.103)	(0.114)		(0.103)	(0.115)		(0.129)	(0.0764)		(0.129)	(0.0766)		(0.129)	(0.0765)
Lagged dependent variable	0.731***	0.731***	0.0841***	0.731***	0.731***	0.0841***	0.731***	0.731***	0.0841***	0.759***	0.758***	0.0653***	0.759***	0.759***	0.0653***	0.759***	0.759***	0.0653***
	(0.00917)	(0.00917)	(0.0143)	(0.00914)	(0.00914)	(0.0143)	(0.00914)	(0.00914)	(0.0143)	(0.00992)	(0.00991)	(0.0150)	(0.00992)	(0.00990)	(0.0149)	(0.00991)	(0.00990)	(0.0150)
Constant	0.626***	0.639***	3.863***	0.353***	0.354***	3.541***	0.437***	0.441***	3.681***	0.581***	0.579***	3.625***	0.324***	0.302***	3.421***	0.310***	0.304***	3.446***
	(0.108)	(0.111)	(0.229)	(0.0444)	(0.0502)	(0.207)	(0.0328)	(0.0335)	(0.202)	(0.133)	(0.133)	(0.203)	(0.0670)	(0.0636)	(0.165)	(0.0306)	(0.0307)	(0.154)
Number of observations	28 626	28 626	27 958	28 626	28 626	27 958	28 626	28 626	27 958	34 767	34 767	34 028	34 767	34 767	34 028	34 767	34 767	34 028
Number of individuals	4727	4727	4059	4727	4727	4059	4727	4727	4059	5539	5539	4800	5539	5539	4800	5539	5539	4800
Adjusted R-squared	0.54	0.54	0.71	0.54	0.54	0.71	0.54	0.54	0.71	0.60	0.60	0.76	0.60	0.60	0.76	0.60	0.60	0.76
Year-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y
Panel B: College education	**No college education**									**With college education**								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.0402	−0.0477*	−0.0863***							−0.0411**	−0.0412**	−0.00878						
	(0.0264)	(0.0257)	(0.0327)							(0.0204)	(0.0205)	(0.0199)						
Log JU				−0.0777	−0.0972	−0.175							−0.0733	−0.125*	−0.192**			
				(0.0984)	(0.104)	(0.176)							(0.0613)	(0.0698)	(0.0927)			
Log Disagreement							−0.0475	−0.0500	−0.0510							0.000781	−0.00431	0.00711
							(0.0347)	(0.0363)	(0.0444)							(0.0151)	(0.0171)	(0.0209)
Lagged real-time GDP growth		0.00117***	0.00430***		0.000915**	0.00392**		0.000778	0.00389**		−0.000737***	−0.000354		−0.00111***	−0.000958*		−0.000685*	−0.000254
		(0.000337)	(0.00138)		(0.000443)	(0.00152)		(0.000566)	(0.00156)		(0.000269)	(0.000522)		(0.000417)	(0.000535)		(0.000377)	(0.000601)
Lagged real-time employment rate		−0.00185***	−0.000740		−0.00182**	−0.000634		−0.00183***	−0.000633		0.000148	0.000128		0.000101	−0.0000955		0.000287	0.000193
		(0.000690)	(0.000948)		(0.000746)	(0.000975)		(0.000675)	(0.00100)		(0.000258)	(0.000345)		(0.000315)	(0.000323)		(0.000254)	(0.000387)
Being unemployed		0.177	−0.0649		0.179	−0.0618		0.181	−0.0616		−0.0435	−0.00108		−0.0430	0.00183		−0.0440	−0.00214
		(0.113)	(0.119)		(0.113)	(0.119)		(0.112)	(0.118)		(0.110)	(0.0820)		(0.110)	(0.0821)		(0.110)	(0.0817)
Lagged dependent variable	0.760***	0.760***	0.0724***	0.760***	0.760***	0.0727***	0.760***	0.760***	0.0727***	0.738***	0.738***	0.0713***	0.738***	0.738***	0.0712***	0.738***	0.738***	0.0713***
	(0.0105)	(0.0105)	(0.0154)	(0.0105)	(0.0105)	(0.0154)	(0.0105)	(0.0105)	(0.0154)	(0.00733)	(0.00732)	(0.0128)	(0.00733)	(0.00732)	(0.0127)	(0.00732)	(0.00732)	(0.0128)
Constant	0.524***	0.558***	4.678***	0.277***	0.264***	4.155***	0.283***	0.279***	4.190***	0.606***	0.608***	3.351***	0.358***	0.330***	3.215***	0.402***	0.399***	3.315***
	(0.137)	(0.133)	(0.260)	(0.0611)	(0.0644)	(0.222)	(0.0368)	(0.0377)	(0.210)	(0.108)	(0.107)	(0.180)	(0.0436)	(0.0469)	(0.146)	(0.0313)	(0.0322)	(0.146)
Number of observations	22 918	22 918	22 227	22 918	22 918	22 227	22 918	22 918	22 227	42 965	42 965	42 123	42 965	42 965	42 123	42 965	42 965	42 123
Number of individuals	4083	4083	3392	4083	4083	3392	4083	4083	3392	6786	6786	5944	6786	6786	5944	6786	6786	5944
Adjusted R-squared	0.60	0.60	0.76	0.60	0.60	0.76	0.60	0.60	0.76	0.56	0.56	0.72	0.56	0.56	0.72	0.56	0.56	0.72
Year-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

on smooth ambiguity preferences, the ambiguity-adjusted expected outcome in scenario k is given by:

$$\phi(\nu(E^k(Y))) = \exp\left(-\text{AA} \cdot 100 \cdot \nu(E^k(Y))\right), \tag{8}$$

where AA ≥ 0 is the ambiguity aversion parameter.

Given $\phi(\cdot)$, the subjective probability of the worst-case scenario is:

$$P(w) = \frac{\mu\, \phi(\nu(E^w(Y)))}{\mu\, \phi(\nu(E^w(Y))) + (1-\mu)\, \phi\left(\nu\left(E^b(Y)\right)\right)}. \tag{9}$$

The overall expected outcome and its standard deviation are:

$$E(Y) = P(w)\, \nu(E^w(Y)) + [1 - P(w)]\, \nu\left(E^b(Y)\right), \tag{10}$$

$$\sigma(Y) = \Bigg[P(w)\left(\sigma_w^2 + [E(Y) - \nu(E^w(Y))]^2\right) + [1-P(w)]\left(\sigma_b^2 + \left[E(Y) - \nu\left(E^b(Y)\right)\right]^2\right)\Bigg]^{0.5} \tag{11}$$

(1) Risk simulation

For the risk simulation:

- Ambiguity aversion is fixed at AA = 0.
- Risk R is varied from 0 to 0.2.
- The standard deviations are set to:

$$\sigma_w = \sigma_b = R.$$

- The expected outcomes are fixed at:

$$E^b(Y) = 0.05, \quad E^w(Y) = E^b(Y) - a_{\text{const}},$$
$$a_{\text{const}} = 0.075 \ (\Rightarrow E^w(Y) = -0.025).$$

- The coefficient of risk aversion RA takes the values {0, 0.5, 1.0}.

For each value of R and RA, we compute $\nu(E^k(Y))$ from (7), $P(w)$ from (9), and then $E(Y)$ and $\sigma(Y)$ from (10)–(11).

(2) Ambiguity simulation

For the ambiguity simulation:

- Risk aversion is fixed at RA = 0.
- Ambiguity level A is varied from 0 to 0.2.

Table 17
Households subjective uncertainty (GBD) by personal instability.

Panel A: Job tenure	Main job < 5 Years									Main job ≥ 5 Years								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18
Log EPU	0.0340	0.0321	−0.00218							−0.0654***	−0.0650***	−0.0381*						
	(0.0256)	(0.0262)	(0.0251)							(0.0217)	(0.0220)	(0.0215)						
Log JU				0.0208	0.0353	−0.200							−0.0190	−0.0744	−0.0300			
				(0.0796)	(0.101)	(0.139)							(0.0971)	(0.0867)	(0.105)			
Log Disagreement							−0.0392*	−0.0418*	0.0248							−0.00677	−0.0172	0.0130
							(0.0200)	(0.0223)	(0.0294)							(0.0151)	(0.0169)	(0.0302)
Lagged real-time GDP growth		0.000666	0.00155*		0.000723	0.000903		0.000175	0.00184*		−0.00122*	0.000842		−0.00135*	0.000828		−0.00124	0.00108
		(0.000471)	(0.000895)		(0.000648)	(0.000884)		(0.000614)	(0.00102)		(0.000689)	(0.000786)		(0.000775)	(0.000833)		(0.000819)	(0.000908)
Lagged real-time employment rate		−0.000416	−0.000144		−0.000482	−0.000405		−0.000675	−0.0000248		0.000506	0.000444		0.000627	0.000551		0.000694*	0.000653
		(0.000518)	(0.000498)		(0.000613)	(0.000480)		(0.000499)	(0.000606)		(0.000354)	(0.000429)		(0.000442)	(0.000480)		(0.000404)	(0.000515)
Being unemployed		0.120	−0.0669		0.120	−0.0632		0.124	−0.0696		0.111	−0.0172		0.112	−0.0173		0.111	−0.0189
		(0.106)	(0.139)		(0.107)	(0.139)		(0.106)	(0.139)		(0.135)	(0.0790)		(0.135)	(0.0787)		(0.135)	(0.0788)
Lagged dependent variable	0.685***	0.684***	0.0614***	0.685***	0.684***	0.0613***	0.685***	0.684***	0.0614***	0.743***	0.743***	0.0397*	0.743***	0.743***	0.0397*	0.743***	0.743***	0.0397*
	(0.0171)	(0.0171)	(0.0175)	(0.0171)	(0.0171)	(0.0175)	(0.0171)	(0.0171)	(0.0175)	(0.0175)	(0.0174)	(0.0235)	(0.0175)	(0.0174)	(0.0234)	(0.0175)	(0.0174)	(0.0235)
Constant	0.288**	0.296**	3.045***	0.470***	0.477***	2.938***	0.423***	0.421***	3.063***	0.606***	0.606***	3.051***	0.270***	0.239***	2.846***	0.275***	0.269***	2.875***
	(0.129)	(0.131)	(0.213)	(0.0611)	(0.0712)	(0.197)	(0.0518)	(0.0528)	(0.189)	(0.112)	(0.114)	(0.185)	(0.0722)	(0.0641)	(0.164)	(0.0405)	(0.0400)	(0.154)
Number of observations	29 880	29 880	29 154	29 880	29 880	29 154	29 880	29 880	29 154	37 545	37 545	36 687	37 545	37 545	36 687	37 545	37 545	36 687
Number of individuals	4999	4999	4273	4999	4999	4273	4999	4999	4273	6077	6077	5219	6077	6077	5219	6077	6077	5219
Adjusted R-squared	0.48	0.48	0.66	0.48	0.48	0.66	0.48	0.48	0.66	0.58	0.58	0.75	0.58	0.58	0.75	0.58	0.58	0.75
Year-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Panel B: College education	No college education									With college education								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.0159	−0.0180	−0.0575**							−0.0142	−0.0151	0.00644						
	(0.0291)	(0.0303)	(0.0289)							(0.0172)	(0.0174)	(0.0182)						
Log JU				0.0298	0.0279	0.00284							−0.0175	−0.0609	−0.135			
				(0.0918)	(0.111)	(0.156)							(0.0660)	(0.0848)	(0.0993)			
Log Disagreement							−0.0569**	−0.0637***	−0.0314							−0.00656	−0.0139	0.0357
							(0.0226)	(0.0241)	(0.0432)							(0.0158)	(0.0182)	(0.0224)
Lagged real-time GDP growth		0.000210	0.00433***		0.000358	0.00446***		−0.000382	0.00409***		−0.000610**	−0.000124		−0.000802*	−0.000573		−0.000711*	0.000274
		(0.000913)	(0.00118)		(0.00100)	(0.00127)		(0.000828)	(0.00134)		(0.000286)	(0.000545)		(0.000420)	(0.000592)		(0.000369)	(0.000642)
Lagged real-time employment rate		−0.000405	0.000213		−0.000291	0.000451		−0.000548	0.000297		0.0000476	0.0000775		0.00000692	−0.000128		0.0000618	0.000212
		(0.000921)	(0.000770)		(0.000970)	(0.000847)		(0.000820)	(0.000826)		(0.000335)	(0.000339)		(0.000405)	(0.000336)		(0.000350)	(0.000422)
Being unemployed		0.107	−0.0128		0.106	−0.0150		0.111	−0.0114		0.107	−0.0156		0.107	−0.0133		0.108	−0.0192
		(0.0887)	(0.137)		(0.0895)	(0.137)		(0.0884)	(0.137)		(0.151)	(0.0803)		(0.151)	(0.0805)		(0.151)	(0.0800)
Lagged dependent variable	0.756***	0.756***	0.0525**	0.756***	0.756***	0.0525**	0.756***	0.756***	0.0525**	0.671***	0.671***	0.0397**	0.671***	0.671***	0.0396**	0.671***	0.671***	0.0397**
	(0.0161)	(0.0161)	(0.0217)	(0.0161)	(0.0161)	(0.0217)	(0.0161)	(0.0161)	(0.0217)	(0.0149)	(0.0148)	(0.0174)	(0.0149)	(0.0148)	(0.0173)	(0.0149)	(0.0148)	(0.0174)
Constant	0.399***	0.408***	3.848***	0.337***	0.335***	3.561***	0.270***	0.265***	3.526***	0.479***	0.485***	2.694***	0.398***	0.374***	2.662***	0.403***	0.398***	2.768***
	(0.143)	(0.150)	(0.249)	(0.0690)	(0.0784)	(0.222)	(0.0428)	(0.0432)	(0.213)	(0.0905)	(0.0914)	(0.169)	(0.0476)	(0.0564)	(0.149)	(0.0374)	(0.0387)	(0.144)
Number of observations	25 224	25 224	24 422	25 224	25 224	24 422	25 224	25 224	24 422	44 959	44 959	44 029	44 959	44 959	44 029	44 959	44 959	44 029
Number of individuals	4565	4565	3763	4565	4565	3763	4565	4565	3763	7184	7184	6254	7184	7184	6254	7184	7184	6254
Adjusted R-squared	0.60	0.60	0.76	0.60	0.60	0.76	0.60	0.60	0.76	0.46	0.46	0.66	0.46	0.46	0.66	0.46	0.46	0.66
Year-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 18
Mean expectations and subjective uncertainty against the daily Economic Policy Uncertainty index.

Panel A: Mean expcetations	directly provided by respondents				mass at mid point method				generalized beta distribution			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Log Daily EPU	−0.0481***	−0.0452***	−0.0540***	−0.0394***	−0.0183	−0.0149	−0.0402**	−0.0213	−0.0235	−0.0200	−0.0431**	−0.0261
	(0.0173)	(0.0171)	(0.0165)	(0.0147)	(0.0182)	(0.0181)	(0.0176)	(0.0144)	(0.0206)	(0.0205)	(0.0187)	(0.0165)
Log average Daily EPU in past 30 days				−0.112**				−0.145***				−0.130**
				(0.0441)				(0.0539)				(0.0567)
Lagged real-time GDP growth		0.00121	0.00290**	0.00213*		0.00172***	0.00480***	0.00380***		0.00222***	0.00577***	0.00487***
		(0.000901)	(0.00124)	(0.00124)		(0.000482)	(0.00139)	(0.00129)		(0.000541)	(0.00148)	(0.00142)
Lagged real-time employment rate		0.00129***	0.00206**	0.00137		0.00121***	0.00134	0.000448		0.00111**	0.00141	0.000608
		(0.000378)	(0.00102)	(0.000904)		(0.000377)	(0.000981)	(0.000855)		(0.000543)	(0.00108)	(0.00100)
Being unemployed		−0.261**	−0.404**	−0.395***		−0.361***	−0.311**	−0.299**		−0.277*	−0.255	−0.244
		(0.110)	(0.146)	(0.146)		(0.0926)	(0.140)	(0.140)		(0.152)	(0.159)	(0.159)
Lagged dependent variable	0.632***	0.632***	0.637***	0.0635***	0.603***	0.603***	0.0370***	0.0367***	0.532***	0.532***	0.00463	0.00447
	(0.00585)	(0.00584)	(0.00852)	(0.00851)	(0.00646)	(0.00646)	(0.00887)	(0.00887)	(0.0100)	(0.0100)	(0.0131)	(0.0131)
Constant	1.143***	1.128***	4.944***	5.431***	1.065***	1.048***	4.088***	4.722***	1.239***	1.219***	4.369***	4.937***
	(0.0890)	(0.0882)	(0.268)	(0.342)	(0.0941)	(0.0932)	(0.259)	(0.374)	(0.108)	(0.108)	(0.291)	(0.403)
Number of observations	70 445	70 445	68 702	68 702	70 387	70 387	68 652	68 652	70 183	70 183	68 451	68 451
Number of individuals	11 780	11 780	10 037	10 037	11 766	11 766	10 031	10 031	11 749	11 749	10 017	10 017
Adjusted R-squared	0.43	0.43	0.60	0.60	0.38	0.38	0.58	0.58	0.30	0.30	0.52	0.52
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y

Panel B: Subjective uncertainty	mass at mid point method				generalized beta distribution				expected ranked probability score			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Log Daily EPU	−0.0204**	−0.0211**	−0.0189**	−0.0213***	−0.0174**	−0.0181**	−0.0143*	−0.0185**	−0.265*	−0.278**	−0.249*	−0.278**
	(0.00844)	(0.00843)	(0.00727)	(0.00757)	(0.00825)	(0.00833)	(0.00748)	(0.00787)	(0.135)	(0.136)	(0.137)	(0.135)
Log average Daily EPU in past 30 days				0.0176				0.0324**				0.222
				(0.0178)				(0.0155)				(0.345)
Lagged real-time GDP growth		−0.000182	0.00100	0.00113*		−0.000432	0.00120**	0.00143**		−0.0128	0.000123	0.00166
		(0.000260)	(0.000628)	(0.000654)		(0.000286)	(0.000595)	(0.000617)		(0.00964)	(0.0118)	(0.0124)
Lagged real-time employment rate		−0.000423*	−0.0000879	0.0000202		−0.000125	0.000119	0.000318		−0.00424	−0.00458	−0.00321
		(0.000219)	(0.000436)	(0.000488)		(0.000274)	(0.000359)	(0.000422)		(0.00389)	(0.00597)	(0.00669)
Being unemployed		0.0596	−0.0272	−0.0287		0.105	−0.0124	−0.0150		−0.419	−0.528	−0.547
		(0.0807)	(0.0710)	(0.0711)		(0.0848)	(0.0772)	(0.0773)		(1.246)	(1.244)	(1.245)
Lagged dependent variable	0.749***	0.749***	0.0732***	0.0732***	0.720***	0.720***	0.0464***	0.0464***	0.664***	0.664***	0.0430***	0.0431***
	(0.00650)	(0.00650)	(0.0107)	(0.0107)	(0.0118)	(0.0118)	(0.0146)	(0.0146)	(0.00454)	(0.00455)	(0.00703)	(0.00702)
Constant	0.464***	0.467***	3.728***	3.652***	0.429***	0.433***	3.097***	2.955***	14.64***	14.73***	66.81***	65.84***
	(0.0419)	(0.0418)	(0.123)	(0.147)	(0.0411)	(0.0415)	(0.125)	(0.132)	(0.845)	(0.856)	(2.332)	(2.979)
Number of observations	65 883	65 883	64 350	64 350	70 183	70 183	68 451	68 451	65 883	65 883	64 350	64 350
Number of individuals	10 869	10 869	9336	9336	11 749	11 749	10 017	10 017	10 869	10 869	9336	9336
Adjusted R-squared	0.58	0.58	0.74	0.74	0.54	0.54	0.72	0.72	0.44	0.44	0.63	0.63
Month-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y

Panel C: Subjective uncertainty	Span between upper- and lower-bound forecasts			
	(1)	(2)	(3)	(4)
Log Daily EPU	−0.0456	−0.0469	−0.0479*	−0.0548*
	(0.0341)	(0.0341)	(0.0277)	(0.0294)
Log average Daily EPU in past 30 days				0.0532
				(0.0753)
Lagged real-time GDP growth		0.000290	0.00362	0.00399
		(0.00104)	(0.00267)	(0.00276)
Lagged real-time employment rate		−0.00119	−0.0000864	0.000239
		(0.00105)	(0.00161)	(0.00179)
Being unemployed		0.110	0.120	0.115
		(0.251)	(0.260)	(0.261)
Lagged dependent variable	0.769***	0.769***	0.0914***	0.0914***
	(0.00563)	(0.00563)	(0.0103)	(0.0103)
Constant	1.330***	1.335***	14.63***	14.39***
	(0.170)	(0.170)	(0.497)	(0.575)
Number of observations	70 765	70 765	69 027	69 027
Number of individuals	11 795	11 795	10 057	10 057
Adjusted R-squared	0.61	0.61	0.76	0.76
Month-Year FE	Y	Y	Y	Y
Individual FE	N	N	Y	Y

Note: The standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *DailyEPU*$_t$ is the Economic Policy Uncertainty index by Baker et al. (2016) on the daily basis.

- The ambiguity level enters via the outcome in the worst-case scenario:

$$E^w(Y) = E^b(Y) - A, \quad E^b(Y) = 0.05.$$

- The standard deviations are fixed at:

$$\sigma_w = \sigma_b = 0.02.$$

- The coefficient of ambiguity aversion AA takes the values {0, 0.5, 1.0}.

Table 19
The shape of forecast histogram of professional forecasters.

	Lower-bound forecast (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	Upper-bound forecast (10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	−0.651***	−0.681***	−0.707***							−0.399	−0.347	−0.424**						
	(0.169)	(0.178)	(0.164)							(0.242)	(0.251)	(0.206)						
Log JU				−1.189*	−1.460*	−1.656**							0.809	1.336	0.561			
				(0.705)	(0.773)	(0.690)							(1.495)	(1.626)	(1.324)			
Log Disagreement							−0.494**	−0.564**	−0.550***							0.180	0.311	0.0439
							(0.208)	(0.217)	(0.202)							(0.342)	(0.340)	(0.283)
Lagged real-time GDP growth		−0.0116	−0.00586		−0.0137	−0.00856		−0.0130	−0.00710		0.0205	0.0237		0.0264	0.0273		0.0242	0.0257
		(0.00982)	(0.00782)		(0.0103)	(0.00821)		(0.0101)	(0.00825)		(0.0329)	(0.0248)		(0.0307)	(0.0236)		(0.0312)	(0.0239)
Lagged dependent variable	0.788***	0.790***	0.547***	0.787***	0.789***	0.546***	0.788***	0.790***	0.548***	0.712***	0.711***	0.536***	0.717***	0.717***	0.542***	0.715***	0.716***	0.540***
	(0.0281)	(0.0272)	(0.0487)	(0.0283)	(0.0274)	(0.0495)	(0.0282)	(0.0273)	(0.0490)	(0.0445)	(0.0444)	(0.0602)	(0.0410)	(0.0398)	(0.0564)	(0.0426)	(0.0417)	(0.0580)
Survey since 2020Q1	−0.0246	−0.0636	0.119	0.0253	−0.0463	0.121	0.872*	0.937**	1.104**	3.652**	3.724**	4.834***	3.862***	3.995***	5.015***	3.509***	3.385***	4.893***
	(0.396)	(0.317)	(0.336)	(0.424)	(0.300)	(0.302)	(0.454)	(0.401)	(0.445)	(1.472)	(1.442)	(1.206)	(1.468)	(1.396)	(1.175)	(1.098)	(1.157)	(1.047)
Constant	3.131***	3.297***	3.734***	−0.599	−0.717	−0.506	−0.126	−0.122	0.201	3.451***	3.161***	4.246***	1.996**	2.222**	2.528***	1.620***	1.610***	2.248***
	(0.812)	(0.863)	(0.813)	(0.425)	(0.446)	(0.387)	(0.233)	(0.239)	(0.205)	(1.058)	(1.136)	(0.949)	(0.914)	(0.980)	(0.823)	(0.284)	(0.298)	(0.336)
Number of observations	3774	3774	3761	3774	3774	3761	3774	3774	3761	3774	3774	3761	3774	3774	3761	3774	3774	3761
Number of individuals	159	159	146	159	159	146	159	159	146	159	159	146	159	159	146	159	159	146
Adjusted R-squared	0.64	0.64	0.68	0.64	0.64	0.68	0.64	0.64	0.68	0.78	0.78	0.80	0.78	0.78	0.80	0.78	0.78	0.80
Quarter-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

	Span between Upper- and Lower- bound forecasts (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log EPU	0.312*	0.394	0.301						
	(0.178)	(0.354)	(0.267)						
Log JU				2.429***	3.263***	2.363***			
				(0.728)	(0.888)	(0.815)			
Log Disagreement							0.846*	1.058**	0.661**
							(0.433)	(0.455)	(0.331)
Lagged real-time GDP growth		0.0322	0.0299		0.0418**	0.0368**		0.0386	0.0137
		(0.0473)	(0.0336)		(0.0179)	(0.0146)		(0.0452)	(0.0325)
Lagged dependent variable	0.829***	0.830***	0.577***	0.831***	0.833***	0.582***	0.832***	0.833***	0.582***
	(0.0231)	(0.0266)	(0.0707)	(0.0231)	(0.0220)	(0.0666)	(0.0285)	(0.0255)	(0.0699)
Survey since 2020Q1	3.003***	3.103	4.489***	3.225***	3.431***	4.698***	1.715	1.507	3.460***
	(0.514)	(1.946)	(1.532)	(0.513)	(0.526)	(0.585)	(1.455)	(1.545)	(1.292)
Constant	−0.461	−0.931	0.270	2.349***	2.704***	2.966***	1.317***	1.297***	1.910***
	(0.851)	(1.586)	(1.189)	(0.456)	(0.525)	(0.534)	(0.263)	(0.290)	(0.362)
Number of observations	3774	3774	3761	3774	3774	3761	3774	3774	3761
Number of individuals	159	159	146	159	159	146	159	159	146
Adjusted R-squared	0.77	0.77	0.80	0.77	0.77	0.80	0.77	0.77	0.80
Quarter-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). *Disagreement* is the forecast disagreement of the next-year GDP growth by professional forecasters.

Table 20
The shape of forecast histogram of households.

	Lower-bound forecast (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	Upper-bound forecast (10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log EPU	0.0121	0.0335	−0.0172							−0.109**	−0.139***	−0.159***						
	(0.0441)	(0.0536)	(0.0569)							(0.0545)	(0.0458)	(0.0481)						
Log JU				−0.00394	−0.0831	−0.341							−0.0747	−0.691**	−1.244***			
				(0.0851)	(0.152)	(0.292)							(0.133)	(0.289)	(0.332)			
Log Disagreement							0.00637	−0.0569	−0.214***							0.0139	−0.119	−0.286***
							(0.0313)	(0.0499)	(0.0805)							(0.0535)	(0.0826)	(0.0755)
Lagged real-time GDP growth	0.000271	−0.000185	0.00266	0.000219	−0.000572	0.00160	0.000307	−0.000823	0.000240	0.00286	0.000380	0.00634***	0.00293	−0.00188*	0.00268	0.00336	−0.000469	0.00338
	(0.00165)	(0.000832)	(0.00174)	(0.00160)	(0.000951)	(0.00191)	(0.00171)	(0.00117)	(0.00176)	(0.00243)	(0.000598)	(0.00238)	(0.00317)	(0.00101)	(0.00217)	(0.00364)	(0.00113)	(0.00254)
Lagged real-time employment rate	0.00227**	0.00217***	0.00206*	0.00223**	0.00191***	0.00166	0.00224**	0.00187***	0.00116	0.00229	0.000837	0.00176	0.00254	0.000257	0.000695	0.00258	0.000993	0.00109
	(0.00114)	(0.000637)	(0.00107)	(0.00100)	(0.000671)	(0.00102)	(0.00109)	(0.000574)	(0.000914)	(0.00149)	(0.000737)	(0.00191)	(0.00203)	(0.000866)	(0.00157)	(0.00252)	(0.000792)	(0.00154)
Being unemployed	−0.512***	−0.499**	−0.392**	−0.510***	−0.497**	−0.385**	−0.512***	−0.494**	−0.368*	−0.344**	−0.325**	−0.273	−0.351**	−0.315**	−0.250	−0.364**	−0.319**	−0.246
	(0.195)	(0.196)	(0.191)	(0.195)	(0.196)	(0.191)	(0.195)	(0.196)	(0.191)	(0.154)	(0.155)	(0.205)	(0.155)	(0.155)	(0.204)	(0.155)	(0.154)	(0.205)
Lagged dependent variable	0.683***	0.682***	0.0542***	0.683***	0.682***	0.0542***	0.683***	0.682***	0.0540***	0.720***	0.718***	0.0639***	0.720***	0.718***	0.0637***	0.720***	0.718***	0.0636***
	(0.00733)	(0.00735)	(0.00962)	(0.00733)	(0.00735)	(0.00962)	(0.00733)	(0.00735)	(0.00962)	(0.00440)	(0.00443)	(0.00703)	(0.00439)	(0.00444)	(0.00704)	(0.00437)	(0.00443)	(0.00703)
Constant	0.0694	−0.190	−3.674***	0.128***	−0.0716	−3.928***	0.134***	−0.0712	−4.003***	2.238***	2.090***	11.95***	1.662***	0.991***	10.55***	1.699***	1.295***	10.84***
	(0.222)	(0.269)	(0.434)	(0.0376)	(0.0996)	(0.357)	(0.0227)	(0.0744)	(0.333)	(0.280)	(0.227)	(0.439)	(0.0573)	(0.163)	(0.397)	(0.0411)	(0.0850)	(0.377)
Number of observations	70765	70765	69027	70765	70765	69027	70765	70765	69027	70765	70765	69027	70765	70765	69027	70765	70765	69027
Number of individuals	11795	11795	10057	11795	11795	10057	11795	11795	10057	11795	11795	10057	11795	11795	10057	11795	11795	10057
Adjusted R-squared	0.48	0.48	0.67	0.48	0.48	0.67	0.48	0.48	0.67	0.53	0.53	0.69	0.53	0.53	0.69	0.53	0.53	0.69
Year-Month FE	N	Y	Y	N	Y	Y	N	Y	Y	N	Y	Y	N	Y	Y	N	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y	N	N	Y

	Span between Upper- and Lower- bound forecasts (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log EPU	−0.115**	−0.170**	−0.141**						
	(0.0487)	(0.0810)	(0.0683)						
Log JU				−0.0581	−0.555*	−0.893**			
				(0.113)	(0.312)	(0.380)			
Log Disagreement							0.00455	−0.0789	−0.0753
							(0.0464)	(0.0872)	(0.0747)
Lagged real-time GDP growth	0.00218*	0.000154	0.00360	0.00233	−0.00152	0.00102	0.00259	−0.000228	0.00303
	(0.00123)	(0.000923)	(0.00254)	(0.00203)	(0.00145)	(0.00265)	(0.00234)	(0.00146)	(0.00279)
Lagged real-time employment rate	−0.000214	−0.00161	−0.000395	0.0000629	−0.00185	−0.00105	0.0000873	−0.00121	−0.000180
	(0.000776)	(0.00114)	(0.00156)	(0.00118)	(0.00128)	(0.00142)	(0.00145)	(0.00115)	(0.00162)
Being unemployed	0.104	0.110	0.119	0.0937	0.116	0.134	0.0858	0.112	0.123
	(0.249)	(0.251)	(0.260)	(0.250)	(0.251)	(0.260)	(0.249)	(0.250)	(0.260)
Lagged dependent variable	0.769***	0.769***	0.0914***	0.769***	0.769***	0.0913***	0.769***	0.769***	0.0914***
	(0.00563)	(0.00563)	(0.0103)	(0.00563)	(0.00563)	(0.0103)	(0.00563)	(0.00563)	(0.0103)
Constant	1.841***	1.962***	15.10***	1.242***	0.789***	13.96***	1.267***	1.047***	14.31***
	(0.249)	(0.420)	(0.582)	(0.0516)	(0.183)	(0.480)	(0.0395)	(0.114)	(0.479)
Number of observations	70765	70765	69027	70765	70765	69027	70765	70765	69027
Number of individuals	11795	11795	10057	11795	11795	10057	11795	11795	10057
Adjusted R-squared	0.61	0.61	0.76	0.61	0.61	0.76	0.61	0.61	0.76
Month-Year FE	N	Y	Y	N	Y	Y	N	Y	Y
Individual FE	N	N	Y	N	N	Y	N	N	Y

Note: Standard errors in parentheses are clustered at the individual and time levels. The regressions are estimated on pooled data across individuals. *EPU* is the Economic Policy Uncertainty index by Baker et al. (2016). *JU* is the 1-month ahead Macroeconomic Uncertainty Index by Jurado et al. (2015). Disagreement is the forecast disagreement of the next-year GDP growth by professional forecasters.

For each value of A and AA, we compute $\nu(E^k(Y))$ from (7), $P(w)$ from (9), and then $E(Y)$ and $\sigma(Y)$ from (10)–(11).

Data and code availability

The code and data required to reproduce the results in this paper are available at https://github.com/poramapa-p/replication_IFJ2026. Due to file size limitations, processed datasets are hosted externally and can be accessed at https://doi.org/10.5281/zenodo.19449956.

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